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A community-driven recommender system for speed safety camera policy using large language models and knowledge graphs

Farhad Sedighi [a], Brian M. Staes [a], Benyamin Ghoreishi [b], Xuehan Jing [a], Haizhong Wang [a,*]

[a] School of Civil and Environmental Engineering and Earth Sciences, Clemson University, Clemson, SC, 29634, USA
[b] School of Civil and Construction Engineering, Oregon State University, Corvallis, OR, 97331, USA

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ABSTRACT

Speeding is a persistent contributor to roadway fatalities, accounting for over 30% of traffic deaths in the U.S., prompting the widespread use of Speed Safety Cameras (SSCs) as a proven countermeasure. However, despite their effectiveness, public skepticism, which is driven by concerns over fairness, transparency, and revenue motives, has limited the scalability and sustainability of SSC programs. Existing research lacks systematic integration of public perception into policy frameworks and fails to fully leverage advancements in artificial intelligence for context-aware recommendation. Addressing this gap, this study develops a scalable policy recommender system for SSC deployment in Oregon by fusing public sentiment with authoritative guidance using a novel hybrid framework that integrates Large Language Models (LLMs), Knowledge Graphs (KGs), and Retrieval-Augmented Generation (RAG). A statewide survey of 1000 residents captured public opinion across six policy dimensions, which, along with state legislation (HB4109), FHWA guidelines, and best practices, formed a multi-source knowledge base stored in a Neo4j graph. The system retrieves both structured and unstructured context to support LLM-based generation of interpretable policy recommendations. Through Monte Carlo simulations, sentence similarity benchmarking, and RAGAS metrics, the proposed framework achieved the highest performance in accuracy, faithfulness, and context recall compared to baseline LLM and RAG pipelines. The results underscore the effectiveness of combining symbolic reasoning with semantic retrieval for generating transparent and community-aligned SSC policy guidance, offering an adaptable framework for jurisdictions aiming to align traffic safety interventions with public trust and domain expertise.

1. Introduction

Speeding is a major contributor to traffic fatalities and injuries, responsible for over 30% of fatal crashes in the U.S. and billions in economic losses (Rodier et al., 2007; Thomas et al., 2008; Morain et al., 2016; Lyon et al., 2024; King et al., 2024). Speed Safety Cameras (SSCs) have consistently proven effective in reducing speeding, crashes, and fatalities worldwide, with studies reporting injury crash reductions of 20-50% and fatality reductions up to 71% (Pilkington and Kinra, 2005; Thomas et al., 2008; McCarty, 2023; Lyon et al., 2024). This technology is also referred to interchangeably as Automated Speed Enforcement (ASE), with both terms denoting the same underlying system. SSCs offer a reliable, equitable, and cost-effective alternative to traditional enforcement,

* Corresponding author.
E-mail address: haizhow@clemson.edu (H. Wang).

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especially amid police staffing shortages and rising safety concerns (Morain et al., 2016; King et al., 2024). Recognized as a "proven safety countermeasure" by FHWA and NHTSA, SSCs support Vision Zero goals when deployed with clear guidelines, public support, and strategic planning (Federal Highway Administration and National Highway Traffic Safety Administration, 2023; King et al., 2024). This aligns with recent Swedish regional evidence showing that both speed limit reductions and safety cameras significantly reduce fatal and severe crashes, though the magnitude of impact varies across counties and land-use contexts, which highlights the importance of strategic, context-sensitive deployment in advancing Vision Zero outcomes (Zheng et al., 2024).

In light of persistent speed-related fatalities and the demonstrated efficacy of SSCs, providing systematic recommendations on the refinement of speed camera guidelines and related legislative changes is both urgent and necessary. Evidence from Korea and other jurisdictions shows that strategically targeted deployments, informed by accident history, overspeeding data, and expert evaluation, can lead to significant reductions in crash frequency (Onuean et al., 2020). However, when SSC guidelines remain vague or overly qualitative, implementation becomes challenging. Transforming these guidelines into measurable criteria, as demonstrated in Edmonton's example, has been shown to improve deployment decisions (Li et al., 2016). Another study further reinforces this need by showing wide variability in camera effectiveness across sites and jurisdictions, underscoring the necessity for standardized, evidence-based deployment criteria and robust evaluation protocols (Job, 2022).

Public opinion is critical to the success and sustainability of SSC programs. Studies across the U.S., Australia, and Israel show that public support increases when SSC is perceived as fair, transparent, and focused on safety rather than revenue (Soole et al., 2008; Retting et al., 2008; Fleiter and Watson, 2012; Schechtman et al., 2016; Beaton et al., 2022). Concerns over legitimacy and transparency can undermine effectiveness, while targeted outreach and community engagement can improve acceptance, especially in high-risk areas like school zones (Hu and McCartt, 2016; Peterson et al., 2017; Tilahun, 2023). Integrating public perception into SSC policy ensures enforcement strategies are equitable, trusted, and aligned with community values.

To address persistent challenges in the deployment of SSCs, this study aims to develop a scalable, transparent, and community-aligned policy recommendation framework by integrating public opinion, authoritative policy documents, and cutting-edge artificial intelligence methods. Specifically, the goal is to:

1. Embed statewide public perceptions into SSC policy analysis to align enforcement with community trust and legitimacy;
2. Pioneer a novel recommender system that fuses Large Language Models (LLMs), Knowledge Graphs (KGs), and Retrieval-Augmented Generation (RAG) to generate explainable and context-rich recommendations;
3. Demonstrate the system's effectiveness and transferability through rigorous evaluation benchmarks, offering a process-driven and scalable framework, formalized as a generalizable, jurisdiction-agnostic algorithmic procedure, that can be adapted across jurisdictions rather than tied to a fixed set of policy recommendations from a single-state case study.

The proposed system is designed as a decision-support tool rather than an autonomous policy-making system. The outputs generated by the LLM-based framework are advisory and intended to assist transportation agencies in synthesizing complex, multi-source information. Final decisions regarding SSC deployment, enforcement strategies, and legislative changes remain the responsibility of authorized policymakers and agencies. This distinction is critical in the context of transportation policy, where recommendations carry legal and societal implications. The system enhances transparency and interpretability but does not assume accountability for policy implementation.

The remaining sections of the paper are organized as follows: Section two discusses related work, including studies on public perception of SSCs, recent advances in LLM-based recommender systems, and the research gap addressed by this study. The third section provides details about the methodology, including a formal algorithmic specification of the retrieval-generation pipeline. The subsequent section presents the experiments and results of the proposed recommender system, including public opinion survey findings, system architecture, evaluation metrics, and comparative performance against baseline RAG pipelines. The mechanisms for adapting the framework to other jurisdictions, along with the study's limitations, are detailed in Section five. Finally, the study's conclusions are presented in the last section.

2. Related work

This section has been categorized into three parts. The first part includes studies that highlight the importance of public perception in SSC implementation, showing that concerns over fairness and transparency can undermine support. The second part deals with advances in LLM-based recommender systems, where integrating Knowledge Graphs enhances accuracy, explainability, and domain relevance for policy guidance. Finally, the concluding part of this section outlines the research gap addressed by this study and highlights the key contributions it offers.

2.1. Integrating public perception into SSC policy

Despite well-documented safety benefits, the implementation of SSCs in the United States has frequently encountered resistance, often due to public skepticism and lack of perceived legitimacy. Early studies from Australia and the U.S. emphasized how public perceptions, particularly beliefs that SSC is a revenue-generating tool rather than a safety intervention, diminish support for speed cameras (Soole et al., 2008; Fleiter and Watson, 2012; Tilahun, 2023; Abouchacra, 2023). Comparable work on broader transport policy acceptability shows that perceived fairness and effectiveness are critical for public acceptance; for instance, Swedish surveys

found that measures viewed as unfair or ineffective face strong resistance regardless of their technical merits (Eriksson et al., 2008). Australian Auditor-General reports and community surveys revealed that enforcement legitimacy, site transparency, and credible communication significantly shape public attitudes toward SSCs (Fleiter and Watson, 2012; Ebrahim and Nikraz, 2014). Similarly, U.S.-based evaluations, including those in Montgomery County and Chicago, showed substantial speed and crash reductions from SSCs, yet acknowledged persistent opposition due to privacy concerns, fairness, and mistrust in deployment criteria (Hu and McCartt, 2016; Tilahun, 2023). These concerns align with the "credibility," "legitimacy," and "social dilemma" themes identified across jurisdictions (Fleiter and Watson, 2012; Beaton et al., 2022; Abouchacra, 2023). The red-light camera literature further demonstrates that public support hinges on how enforcement attributes such as fine levels, camera numbers, and placement are perceived, with strong sensitivity to privacy, fairness, and cost-effectiveness (Egbendewe-Mondzozo et al., 2010).

A key limitation in existing SSC programs has been the insufficient integration of public perception into deployment and evaluation strategies. Multiple studies, including a national review of U.S. policy attitudes and focus groups in Queensland, emphasize that enforcement without community alignment results in public pushback, reduced program sustainability, and limited political will (Soole et al., 2008; Debinski et al., 2014; Delbosc et al., 2025). This dynamic mirrors findings in related domains such as 20 mph limit compliance, where high stated support does not translate to compliant behavior unless reinforced by strong norms and visible enforcement (Tapp et al., 2016). For example, surveys in Israel and Minnesota confirmed that while many drivers acknowledge the safety benefits of SSCs, support is often conditional on the fairness of implementation and the transparency of siting decisions (Schechtman et al., 2016; Peterson et al., 2017). Moreover, targeted safety messaging that contextualizes speeding as a public health crisis rather than a personal behavior has been shown to shift public attitudes positively, particularly among individuals initially skeptical of SSCs (Peterson et al., 2017; Delbosc et al., 2025). This suggests that data-driven deployment alone is insufficient; public acceptance must be cultivated and incorporated into the policy framework. Related work by Afandizadeh et al. (2024) demonstrates that accessibility-enhancing interventions can generate significant distributional impacts that shape affordability, local economic dynamics, and neighborhood desirability, and these effects must be directly incorporated into policy design. These insights highlight the need for SSC guidelines that integrate both safety evidence and broader equity-sensitive community impacts.

Against this backdrop, the innovation of the current Oregon-focused study lies in explicitly embedding community perspectives, gathered through a 1,000-person state-wide public opinion survey, into the policy analysis and deployment strategy for SSCs. While some prior U.S. efforts have assessed public views post-implementation or as an adjunct to speed reduction metrics (Retting et al., 2008; Hu and McCartt, 2016; Beaton et al., 2022), few have systematically incorporated pre-deployment public attitudes into SSC planning. This approach addresses the longstanding implementation gaps and policy opposition documented in national reviews (Debinski et al., 2014), while aligning with international best practices emphasizing the co-production of traffic safety policy with the public (Fleiter and Watson, 2012; Beaton et al., 2022; Ebrahim and Nikraz, 2014). In line with emerging evidence from Norway identifying distinct public "classes" with different acceptance patterns across multiple transport policies, incorporating local attitudes helps avoid one-size-fits-all approaches and instead supports tailored transportation strategies responsive to community values (Mehdizadeh et al., 2024). By linking public sentiment directly to siting logic, deployment transparency, and revenue allocation guidance, this study seeks to bridge the divide between top-down enforcement and community-informed legitimacy, an area that remains underdeveloped but essential in SSC research and practice.

2.2. LLM-based recommendation advances

Recent studies overwhelmingly demonstrate that LLMs outperform traditional recommendation methods by offering context-aware, personalized, and generative capabilities that prior systems lack. Traditional recommender systems, relying on user-item interaction data and multi-stage pipelines (e.g., candidate generation, ranking), face persistent challenges such as cold-start problems, data sparsity, and limited reasoning or generalization abilities (Lin et al., 2025; Yu et al., 2025; Zhao et al., 2024). In contrast, LLMs offer unified, end-to-end architectures that leverage their language understanding and generation capabilities to deliver accurate and contextually rich recommendations without requiring handcrafted features or static embeddings (Vats et al., 2024; Li et al., 2023; Wang et al., 2024b). Unlike conventional models, LLMs can ingest unstructured data, such as reviews, descriptions, or social media text, and infer user intent with greater nuance (Yu et al., 2025; Tian et al., 2024; Sharifi et al., 2026). Generative frameworks like OpenP5 and MMRec further show that LLMs not only streamline recommendation pipelines but also outperform baseline models on benchmark datasets (Xu et al., 2024b; Tian et al., 2024; Xu et al., 2024a). Moreover, LLMs excel in multi-modal and cross-domain settings, unifying textual and visual modalities into shared latent spaces for more holistic recommendations (Tian et al., 2024; Deldjoo et al., 2024). This shift from discriminative to generative paradigms, supported by surveys and empirical evaluations across domains, marks a fundamental evolution in recommender system design (Bao et al., 2023; Zhao et al., 2024; Saeidian and Singleton, 2025), establishing LLMs as the new standard for intelligent, adaptive, and transparent recommendation strategies.

Recent studies have demonstrated that integrating LLMs with Knowledge Graphs significantly enhances recommendation accuracy and interpretability compared to LLMs alone. LLMs are known for their strong generative and contextual reasoning abilities but often suffer from hallucinations, factual inaccuracies, and limited domain-specific knowledge due to their general-purpose pretraining. Recent studies emphasize that LLMs, while powerful, lack access to structured and evolving domain knowledge, which limits their precision in recommendation settings (Jovanović and Campbell, 2023; Wang et al., 2024a; Qiu et al., 2024). To address this, several frameworks, such as CoLaKG (Cui et al., 2025), COMPASS (Qiu et al., 2024), and LLM-PKG (Wang et al., 2024a), have successfully demonstrated how enriching LLMs with structured KG content leads to improved semantic understanding, user preference modeling, and more explainable recommendations. In personalized education (Abu-Rasheed et al., 2024) and e-commerce (Wang et al., 2024a),

using KGs as contextual scaffolding has proven to reduce hallucinations, improve factual accuracy, and increase user trust, ultimately outperforming LLM-only baselines in real-world deployments.

Beyond the LLM-KG hybrid, an emerging and more powerful paradigm is the combination of RAG with Knowledge Graphs, often referred to as GraphRAG. This approach enables LLMs to retrieve and reason over structured graph data and unstructured texts simultaneously, leading to substantial performance gains (Meng et al., 2025; Afrose, 2024; Zhang et al., 2025). The KERAG_R model (Meng et al., 2025), which incorporates a GraphRAG module with KG-based instruction tuning, achieved up to 14.89% improvement over state-of-the-art LLM-based recommenders like RecRanker. Similarly, in the healthcare domain, the doctor recommender system demonstrated that combining the LangChain RAG with the Neo4j KG setup outperformed standard RAG chains and GPT-4 baselines across multiple ROUGE and BLEU metrics (Afrose, 2024). The GraphRAG survey further validates this superiority by outlining how multihop reasoning, structure-aware integration, and graph-based indexing solve key challenges in conventional RAG systems (Zhang et al., 2025). Building upon this foundation, our study introduces a novel application of a combined RAG and KG paradigm specifically tailored to the traffic policy domain, particularly SSC deployment guidance. This represents an important innovation, as few prior works have used this combination to generate explainable, data-driven recommendations for transportation policymaking.

2.3. Research gap and major contributions

To begin with, this study is the first to integrate statewide public opinion into SSC policy design in Oregon, addressing a key gap in U.S. SSC programs that have historically overlooked community perspectives. While SSCs are effective, public skepticism, which is driven by concerns about legitimacy, transparency, and revenue motives, continues to limit their acceptance and sustainability (Soole et al., 2008; Fleiter and Watson, 2012; Hu and McCartt, 2016; Abouchacra, 2023). Prior research underscores how transparency, credibility, and perceived fairness shape public support (Ebrahim and Nikraz, 2014; Beaton et al., 2022). By incorporating responses from 1000 Oregonians, this study proactively aligns SSC strategies with public sentiment, improving legitimacy and long-term viability.

Second, this study pioneers the integration of LLMs with KGs to develop a domain-specific SSC Recommender System. While LLMs offer strong generative and contextual abilities, they often face challenges like hallucination and limited domain grounding (Jovanović and Campbell, 2023; Wang et al., 2024a; Qiu et al., 2024). By incorporating structured KG content, the system improves factual accuracy and interpretability (Cui et al., 2025; Abu-Rasheed et al., 2024). Built on recent GraphRAG advances that combine RAG with graph-based reasoning for multihop, structure-aware decision-making (Meng et al., 2025; Afrose, 2024; Zhang et al., 2025), existing studies have primarily focused on domains such as healthcare, e-commerce, and general recommendation systems (Afrose, 2024; Wang et al., 2024a). However, limited research has explored whether hybrid KG-RAG systems can support transportation policy recommendation tasks that integrate legislation, operational guidance, and public perception. Moreover, existing GraphRAG evaluations rarely assess recommendation stability and policy alignment using transportation-specific expert benchmarks and community-driven datasets. Addressing these gaps, this work represents one of the first applications of GraphRAG to transportation policy, particularly SSC deployment.

Lastly, this study introduces a scalable and adaptable framework for other state DOTs and policy domains where integrating community input and diverse data sources is essential. By enabling co-produced, context-aware guidance grounded in explainable and structured graph reasoning, the approach balances data-driven insights with community priorities. It is important to note that the Oregon case is presented as a proof-of-concept implementation rather than a source of universally generalizable policy conclusions. The primary contribution lies in the framework architecture, which can be adapted across jurisdictions but remains inherently context-dependent, which requires retraining and reconfiguration using jurisdiction-specific legislation, public input, and institutional conditions, rather than direct transfer of the Oregon-based model.

3. Methodology

Fig. 1 displays the research framework of our paper, which follows a four-step pipeline that integrates RAG with structured knowledge graphs to produce transparent, policy-relevant recommendations. Step 1 (Survey Responses) focuses on data collection through a public survey, gathering responses from 1000 Oregonians to understand public opinion on SSCs. Step 2 (Knowledge Base) involves building a knowledge base, which combines key findings from the survey with authoritative documents including FHWA national guidance, Oregon state legislation (HB4109), and best practices from other jurisdictions. Step 3 (Context Assembly) performs knowledge graph construction and context assembly by applying an LLM Graph Transformer to extract entities and relationships from the knowledge base, forming a Neo4j-based structured graph. This structured knowledge is then combined with unstructured text to form a hybrid context, then all context chunks are converted into vectors using an embedding model. In Step 4 (Prompt Composition), the system carries out prompt generation and LLM inference. When a user submits a query, such as “recommend improvements, additions, or changes to SSC guidance as actionable items for local or state jurisdictions,” the system also embeds the query into a vector using the same embedding model. The system then performs vector similarity matching between the embedded query and the context vectors to retrieve the most relevant context chunks. The system constructs a prompt by embedding the user query along with the previously assembled context and any prior chat history to support continuity in multi-turn conversations. The fully constructed prompt is passed to an LLM to generate a grounded and context-aware policy recommendation output.

While the framework is demonstrated using Oregon-specific data, its architecture is inherently modular. Each component, including survey inputs, legislative documents, and knowledge graph entities, can be substituted with jurisdiction-specific data sources. This enables the system to be adapted across different states by reconfiguring the knowledge base, updating retrieval sources, and recali-

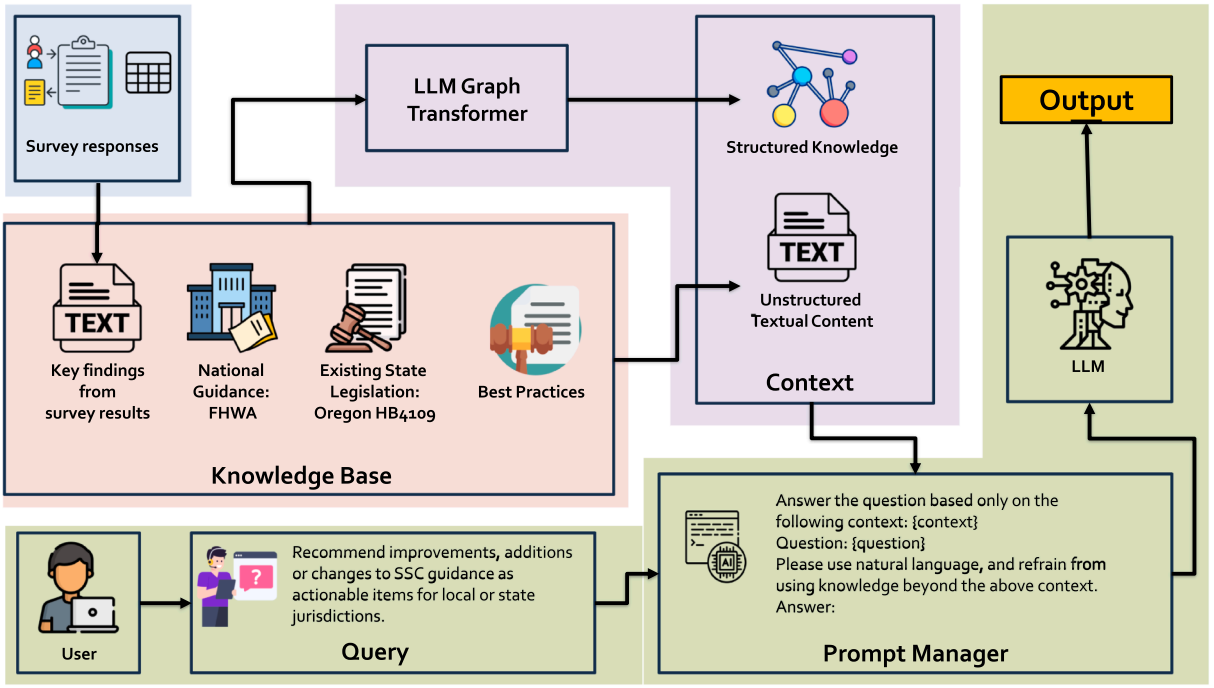


Fig. 1. RAG-KG framework for SSC policy recommendation generation.

brating the model to reflect local regulatory and institutional contexts. The operationalization of this cross-jurisdictional adaptation is further discussed in Section 5.

3.1. Knowledge graphs in recommendation systems

Knowledge Graphs are powerful semantic data structures defined as follows in Eq. (1) (Abu-Rasheed et al., 2024):

$$KG = \{(h, r, t) \mid h, t \in E,\ r \in R\} \tag{1}$$

Where h is the head entity, t is the tail entity, and r is the relation between h and t. E is the entity group and R is the relation group.

Knowledge Graphs represent real-world entities (e.g., people, programs, policies, or zones) as nodes and their relationships (e.g., "USES", "COVERS", or "LOCATED_ON") as edges, enabling machines to understand, query, and reason over complex domains. A key component of KGs is the Property Graph Model, which allows not only labeled nodes and edges but also the inclusion of properties or attributes (e.g., names, types, descriptions) to enrich the graph's informational content. This model supports the representation of both simple and multivalent relationships, such as a vehicle being associated with multiple citations or a citation being linked to multiple entities like registered owners, public agencies, and traffic enforcement agents. Ontologies act as a backbone by defining the schema, that is, the shared concepts and their interrelations, to ensure interoperability and consistency in graph construction (Chen et al., 2024). In practice, platforms like Neo4j manage and query these graphs efficiently using Cypher, a declarative graph query language. The querying process can be represented by Eq. (2):

$$CypherQuery(Pattern, Return) = ResultSet \tag{2}$$

Where a pattern-matching query returns targeted subgraphs or attributes. Such capabilities are essential in domains like recommendation systems, where insight into multi-entity relationships drives personalized, accurate outputs (Miller, 2013). By encoding domain expertise and heterogeneous data sources into a navigable graph, KGs serve as a foundational layer for advanced decision-making systems in transportation, healthcare, and beyond.

3.2. RAG

RAG represents a hybrid methodology that extends the power of LLMs by dynamically retrieving and integrating relevant external information during inference. This architecture overcomes a major limitation of static LLMs by addressing their fixed knowledge cutoff and their inability to access real-time or domain-specific information. RAG is composed of two fundamental components: retrieval, where semantic search or dense passage retrievers identify pertinent documents from a large corpus using vector similarity

techniques, and generation, where the LLM synthesizes a natural language response based on the retrieved context (Zhang et al., 2025). This workflow is captured by Eq. (3):

$$RAG(prompt) = f\big(R(p),\ Understand(R(p)),\ Generate\big(Understand(R(p)),\ prompt\big)\big) \tag{3}$$

where $R(p)$ is the retrieval function applied to the prompt. The Understand and Generate phases refer to internal processing and natural language generation, respectively (Afrose, 2024).

This architecture allows RAG models to adapt responses based on up-to-date and user-specific context, thereby improving relevance, factuality, and interpretability. RAG is especially effective in applications that demand dynamic contextual grounding, such as personalized recommendations, academic research, and multilingual information access (Zhang et al., 2025). Frameworks like LangChain streamline this architecture through modular chains, such as RetrievalQAChain for textual document queries and GraphCypherQAChain for structured KG access via Cypher, thereby enabling high-performance integration between retrieval and generation components in real-world systems (Topsakal and Akinci, 2023).

3.3. Integration of RAG and KG

The integration of Knowledge Graphs, Retrieval-Augmented Generation, and Large Language Models forms a synergistic pipeline that unites structured domain knowledge with unstructured textual reasoning to produce accurate, explainable, and context-rich recommendations. This fusion is powered by embedding models such as OpenAIEmbeddings, which convert textual and graph-based inputs into high-dimensional vector representations. The semantic similarity between two embeddings is computed using Eq. (4):

$$\text{Cosine Similarity} = \frac{A \cdot B}{\|A\|\|B\|} \tag{4}$$

where A and B represent the embedding vectors of two text inputs. The cosine similarity produces a score ranging from -1 to 1: a value of 1 signifies complete similarity, 0 indicates no relation, and -1 denotes complete opposition. In retrieval tasks, only positive similarity scores are considered meaningful, as they reflect semantic relevance. This measure facilitates efficient document and node retrieval based on proximity in semantic space (Reimers and Gurevych, 2019).

Once the relevant knowledge is retrieved, generative models take over, producing coherent and contextually grounded responses. To manage multi-turn conversational logic and customize LLM behavior, LangChain supports modular prompt templates like ChatPromptTemplate, SystemMessagePromptTemplate, and HumanMessagePromptTemplate (Topsakal and Akinci, 2023). However, conversational history does not guarantee that all earlier requirements remain salient as interactions grow longer. LLMs may make premature assumptions or overlook relevant information in long contexts (Laban et al., 2024, 2025). Therefore, multi-turn interaction should be viewed as supporting iterative refinement rather than guaranteeing complete context preservation. In practical workflows, a user query such as “Recommend improvements, additions, or changes to SSC guidance as actionable items for local or state jurisdictions” initiates a RAG pipeline involving a Cypher-based graph query using the KG, retrieval of relevant documents and embeddings, and subsequent prompt construction and LLM response generation. This end-to-end system not only generates fluent and relevant outputs but also grounds them in verifiable knowledge sources. This architecture is especially powerful in recommendation systems across domains like healthcare, intelligent transportation, and transportation policy, where combining symbolic knowledge (from graphs), retrieved evidence, and generative reasoning yields transparent, reliable, and user-aligned outputs. Section 3.4 formalizes this integration as a complete algorithmic procedure spanning Eqs. (1)–(4) above.

3.4. Formal algorithmic specification of the proposed pipeline

To improve the replicability of the proposed framework and to support its comparison against future state-of-the-art systems, this subsection formalizes the methodology as a generalized, jurisdiction-agnostic procedure. The procedure formalizes Steps 2-3 of Fig. 1 (knowledge base and index construction) as an offline phase, and Step 4 (knowledge-guided retrieval-augmented generation) as an online phase.

3.4.1. Offline phase: knowledge base and index construction

Let $\mathcal{D} = \{d_1, d_2, \ldots, d_m\}$ denote the heterogeneous corpus of source documents for a given jurisdiction (e.g., survey summaries, legislative text, national guidelines, and case studies). Each document is decomposed into overlapping token chunks:

$$\mathcal{C} = \bigcup_{i=1}^{m} \text{Split}(d_i;\ \tau,\ \omega), \tag{5}$$

where τ is the chunk size and ω is the overlap length (in tokens), both configurable hyperparameters. The LLM-based graph transformer $\Phi_{\text{LLM}}^{\text{KG}}$ is then applied to every chunk to extract entity-relation triples:

$$KG = \bigcup_{c \in \mathcal{C}} \Phi_{\text{LLM}}^{\text{KG}}(c) = \{(h, r, t) \mid h, t \in E,\ r \in R\}, \tag{6}$$

and persisted in a graph database as (E, R, T), where $T \subseteq E \times R \times E$ is the resulting triple set queryable via Eq. (2). In parallel, each chunk is mapped into a dense vector representation to build the semantic index:

$$\mathcal{V} = \big\{\big(c,\ \mathbf{e}_c\big) \ \big|\ c \in \mathcal{C},\ \mathbf{e}_c = \text{Embed}(c)\big\}. \tag{7}$$

Algorithm 1 presents this offline procedure as generalizable pseudocode. The chunk size τ and overlap ω (e.g., $\tau = 512$ and $\omega = 24$ in this study) are treated as free, configurable parameters.

Algorithm 1 Offline phase: knowledge base and index construction.

Require: Document corpus $D = \{d_1, \dots, d_m\}$; chunk size τ; overlap ω
Ensure: Knowledge graph $KG = (E, R, T)$; vector index $\mathcal{V}$
$\mathcal{C} \leftarrow \emptyset$
for each document $d_i \in D$ **do**
  $\mathcal{C} \leftarrow \mathcal{C} \cup \text{Split}(d_i;\ \tau,\ \omega)$ ▷ Eq. (5)
end for
$KG \leftarrow \emptyset,\ \mathcal{V} \leftarrow \emptyset$
for each chunk $c \in \mathcal{C}$ **do**
  $KG \leftarrow KG \cup \Phi^{\text{KG}}_{\text{LLM}}(c)$ ▷ Entity–relation extraction, Eq. (6)
  $\mathbf{e}_c \leftarrow \text{Embed}(c)$
  $\mathcal{V} \leftarrow \mathcal{V} \cup \{(c, \mathbf{e}_c)\}$ ▷ Eq. (7)
end for
Persist KG in graph database with full-text and fuzzy-match indices
return KG, $\mathcal{V}$

3.4.2. Online phase: knowledge-guided retrieval-augmented generation

At conversational turn t, given a raw user query q_t and prior exchange history $H_{t-1} = \{(q_1, R_1), \dots, (q_{t-1}, R_{t-1})\}$, the system first produces a standalone query:

$$\hat{q}_t = \begin{cases} \text{Condense}(q_t,\ H_{t-1}), & |H_{t-1}| > 0 \\ q_t, & \text{otherwise.} \end{cases} \tag{8}$$

Entities are then extracted from $\hat{q}_t$ using a structured-output LLM call, $\mathcal{Q}_E = \Phi^{\text{ent}}_{\text{LLM}}(\hat{q}_t) = \{e_1, \dots, e_k\}$, and used to drive structured retrieval over the knowledge graph:

$$S(\hat{q}_t) = \bigcup_{j=1}^{k} \text{CypherQuery}\big(\text{Pattern}(e_j,\ \delta),\ \text{Return}\big), \tag{9}$$

where δ denotes an approximate string-matching tolerance for fuzzy entity resolution. Semantic (unstructured) retrieval is performed by embedding $\hat{q}_t$ and ranking all indexed chunks by cosine similarity:

$$U(\hat{q}_t) = \text{Top}_n \left\{ \frac{\mathbf{e}_{\hat{q}_t} \cdot \mathbf{e}_c}{\|\mathbf{e}_{\hat{q}_t}\| \|\mathbf{e}_c\|} \;\middle|\; (c, \mathbf{e}_c) \in \mathcal{V} \right\}, \tag{10}$$

where n is the retrieval depth. The structured and unstructured evidence are then fused into a single hybrid context:

$$\text{Context}(\hat{q}_t) = S(\hat{q}_t) \ \oplus\ U(\hat{q}_t), \tag{11}$$

where $\oplus$ denotes concatenation of the structured and unstructured evidence into a single textual context block. This context is embedded in the prompt template, and the final response is generated by the LLM:

$$P_t = \text{Template}\big(\text{Context}(\hat{q}_t),\ \hat{q}_t\big), \qquad R_t = \text{Generate}\Big(\text{Understand}\big(\text{Context}(\hat{q}_t)\big),\ P_t\Big), \tag{12}$$

with the history updated as $H_t = H_{t-1} \cup \{(q_t, R_t)\}$ to support multi-turn interaction.

Algorithm 2 presents this online procedure as generalizable pseudocode. The retrieval depth n, fuzzy-matching tolerance δ, and generation temperature (e.g., $n = 4$, $\delta = 2$, temperature $= 0$ in this study) are likewise treated as free, configurable parameters.

4. Experiment

The main steps of this section are: first, collecting and analyzing survey responses from 1000 Oregon residents to capture public attitudes toward SSC deployment across six policy-relevant categories; second, constructing a comprehensive knowledge base that integrates these survey insights with authoritative documents and case studies, enabling semantic chunking and knowledge graph development; third, designing a hybrid SSC recommender system that fuses LLMs, structured knowledge graphs, and vector-based retrieval within a RAG framework; fourth, retrieving both structured and unstructured information through dual semantic-symbolic indexing for policy generation; and finally, evaluating system performance using cosine similarity benchmarking via Monte Carlo simulation and RAGAS metrics.

4.1. Survey results

The survey was conducted using Qualtrics, an online panel-based platform, where 1000 respondents were selected based on predefined zip code tabulation areas (ZCTAs) and socio-demographic quotas (Figliozzi et al., 2025). The use of ZCTAs ensured statewide geographic representation, as shown in Fig. 2, while socio-demographic quotas, developed using data available from the

Algorithm 2 Online phase: knowledge-guided retrieval-augmented generation.

Require: Query q_t; history H_{t-1}; KG; $\mathcal{V}$; retrieval depth n; fuzzy tolerance δ
Ensure: Response R_t; updated history H_t

if $|H_{t-1}| > 0$ **then**
  $\hat{q}_t \leftarrow \text{Condense}(q_t, H_{t-1})$ ▷ Eq. (8)
else
  $\hat{q}_t \leftarrow q_t$
end if
$\mathcal{Q}_E \leftarrow \Phi^{\text{ent}}_{\text{LLM}}(\hat{q}_t)$ ▷ Entity extraction from query
$S \leftarrow \emptyset$
for each entity $e_j \in \mathcal{Q}_E$ **do**
  $S \leftarrow S \cup \text{CypherQuery}(\text{Pattern}(e_j, \delta),\ \text{Return})$ ▷ Eq. (9)
end for
$\mathbf{e}_{\hat{q}_t} \leftarrow \text{Embed}(\hat{q}_t)$
$U \leftarrow \text{Top}_n \left\{ \text{CosineSim}(\mathbf{e}_{\hat{q}_t}, \mathbf{e}_c) \mid (c, \mathbf{e}_c) \in \mathcal{V} \right\}$ ▷ Eq. (10)
$\text{Context} \leftarrow S \oplus U$ ▷ Eq. (11)
$P_t \leftarrow \text{Template}(\text{Context},\ \hat{q}_t)$
$R_t \leftarrow \text{Generate}(\text{Understand}(\text{Context}),\ P_t)$ ▷ Eq. (12)
$H_t \leftarrow H_{t-1} \cup \{(q_t, R_t)\}$
return R_t, H_t

U.S. Census Bureau's American Community Survey, ensured representation across key dimensions such as age, gender, race, income, and education.

Representativeness was further strengthened through ZCTA-based geographic stratification, targeted oversampling of underrepresented groups, and post-survey weighting to better align the sample with statewide population distributions. While panel-based surveys enable efficient and scalable data collection, they may introduce potential biases, including sampling bias (due to non-random panel participation) and coverage bias (due to the exclusion of individuals without internet access). To mitigate these limitations, this study employed a combination of quota sampling, stratified sampling strategies, and post-survey weighting. The weighting was applied after the completion of the survey in order to create a more representative sample of the population of Oregon (Figliozzi et al., 2025).

An overview of respondent demographics is presented in Fig. 3. It shows percentage distributions by gender, age, birthplace, race, education, and income. Overall, the sample is relatively balanced by gender, skewed toward middle-aged and older respondents, predominantly U.S.-born, and largely White/Caucasian. Respondents tend to have relatively high levels of education, with many holding bachelor's or graduate degrees, and a broad spread of household incomes.

This section summarizes survey questions that gauge public attitudes across key policy dimensions of SSC deployment, grouped into categories such as expected outcomes, location guidance, transparency, ticketing, and revenue. These insights help agencies design SSC systems with stronger public support. Key themes from each category are outlined below, with full textual analyses integrated into the system's knowledge base.

4.1.1. Expected outcomes

This category examined Oregonians' perceptions of SSC effectiveness. While opinions were mixed regarding broader or indirect outcomes, Fig. 4 indicates strong agreement that SSCs improve roadway safety, particularly by reducing injuries and enhancing conditions for walking and cycling. In contrast, understanding of the halo effect, such as whether speeding increases outside camera zones, appears limited, with responses split between agreement, disagreement, and neutrality. These results suggest that agencies should improve public communication and consider strategies such as dynamic camera placement to extend safety benefits beyond individual corridors.

4.1.2. Location guidance

This category examined public preferences for SSC placement. While the final question reflected limited consensus on the overall presence of SSCs, Fig. 5 shows strong support for deploying cameras in specific, safety-sensitive locations such as school zones, construction zones, community nominated sites, and areas where police enforcement is difficult. These findings suggest agencies should prioritize targeted, data-driven placement strategies that align with public expectations and involve local communities. However, to avoid limiting impacts to isolated corridors, agencies should also consider dynamic deployment approaches to mitigate potential halo effects.

These location preferences are broadly consistent with prior findings from other jurisdictions. In Queensland, participants expressed a clear preference for camera-based enforcement in locations perceived as high-risk, especially accident black spots and school zones, while showing weaker support for camera deployment in areas not viewed as safety-critical (Soole et al., 2008). Similarly, U.S. and Canadian evidence suggests that public acceptance is higher when SSC is linked to visible safety problems, transparent siting rationale, and community protection rather than broad or indiscriminate deployment (Retting et al., 2008; Beaton et al., 2022).

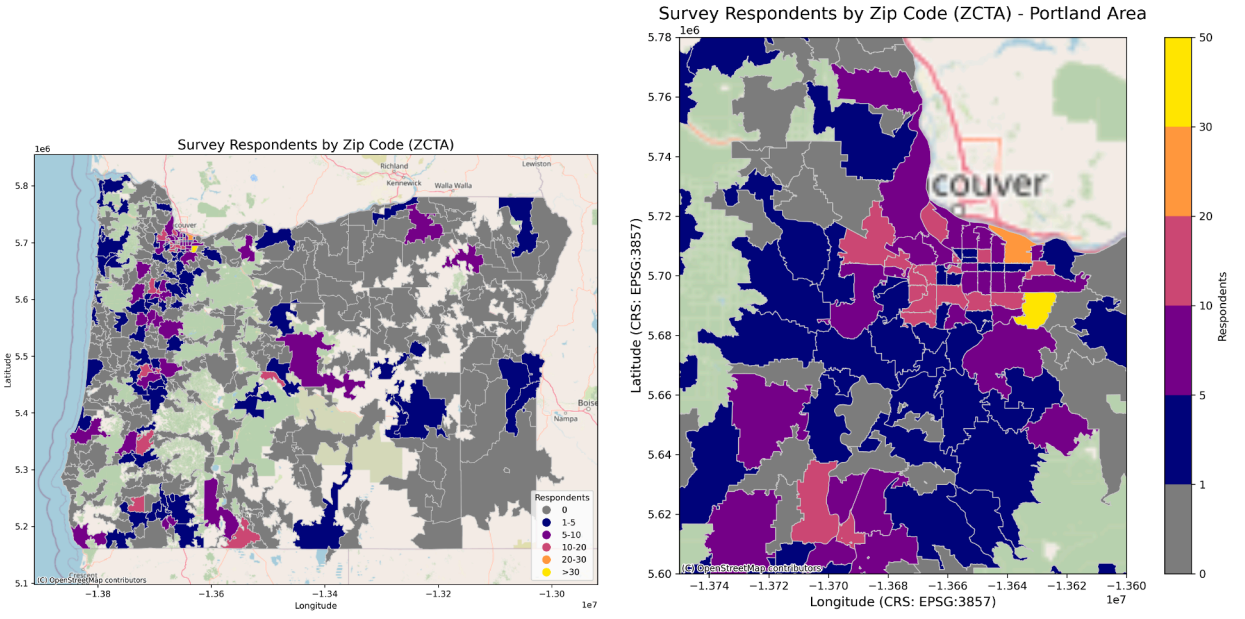


Fig. 2. Survey respondent distribution by ZCTA: (a) statewide distribution across Oregon and (b) detailed distribution within the Portland metropolitan area.

4.1.3. Existing conditions and beliefs

This category examined perceptions of speeding severity, privacy, and enforcement practices. Fig. 6 shows broad agreement that speeding is a problem at both neighborhood and citywide scales, underscoring the applicability of SSCs on local streets and higher-speed corridors. While opinions remain divided on whether SSCs pose privacy concerns in the absence of facial recognition, respondents expressed strong support for mobile or hidden SSC deployments. These adaptive approaches may help mitigate the halo effect while improving speed compliance across the broader roadway network.

4.1.4. Transparency and accountability

This category examined public expectations for transparency in SSC operations. As shown in Fig. 7, respondents strongly support regular reporting of safety outcomes, disclosure of citation and revenue use, and advance public notice of camera locations. These findings highlight that openness and accountability are critical to building trust and securing public support, and should be treated as essential components of any agency's SSC operational strategy.

4.1.5. Ticketing

This category explored public preferences regarding SSC citation design, including fine levels, speed thresholds, and equity considerations. Fig. 8 indicates strong support for higher fines in select high-risk environments, such as school and work zones, while maintaining moderate speed thresholds rather than lowering them. Educational courses were widely endorsed as an alternative to a first citation. Respondents also favored issuing citations to drivers instead of vehicle owners, though there was limited acceptance of facial recognition for driver identification, suggesting a need for further evaluation by agencies. Fig. 9a and b also show that, across both local and state roads, respondents opposed tickets at 5 mph above the posted speed limit and showed the strongest support at 15 to 20 mph above the posted speed limit, indicating a clear preference for targeting more substantial speeding rather than minor exceedances.

The Oregon respondents' stronger support for penalizing more substantial speeding, rather than minor exceedances, is consistent with prior work showing that acceptance of SSC depends heavily on perceptions of fairness and proportionality. In Israel, enforcement was identified as the primary stated reason for compliance with posted speed limits, underscoring the importance of a credible sanction structure (Schechtman et al., 2016). More broadly, recent U.S. evidence suggests that support for SSC increases when it is framed as a response to the safety risks of speeding rather than as a punitive mechanism for low-level violations (Delbosc et al., 2025).

4.1.6. Revenue

This category examined public views on SSC revenues and their appropriate use. As shown in Fig. 10, respondents strongly supported reinvesting SSC funds in roadway infrastructure repair, safety improvements on high-speed or high-crash corridors, policing, and walking and cycling enhancements. While there was broad agreement that SSC programs generate net revenue, respondents did

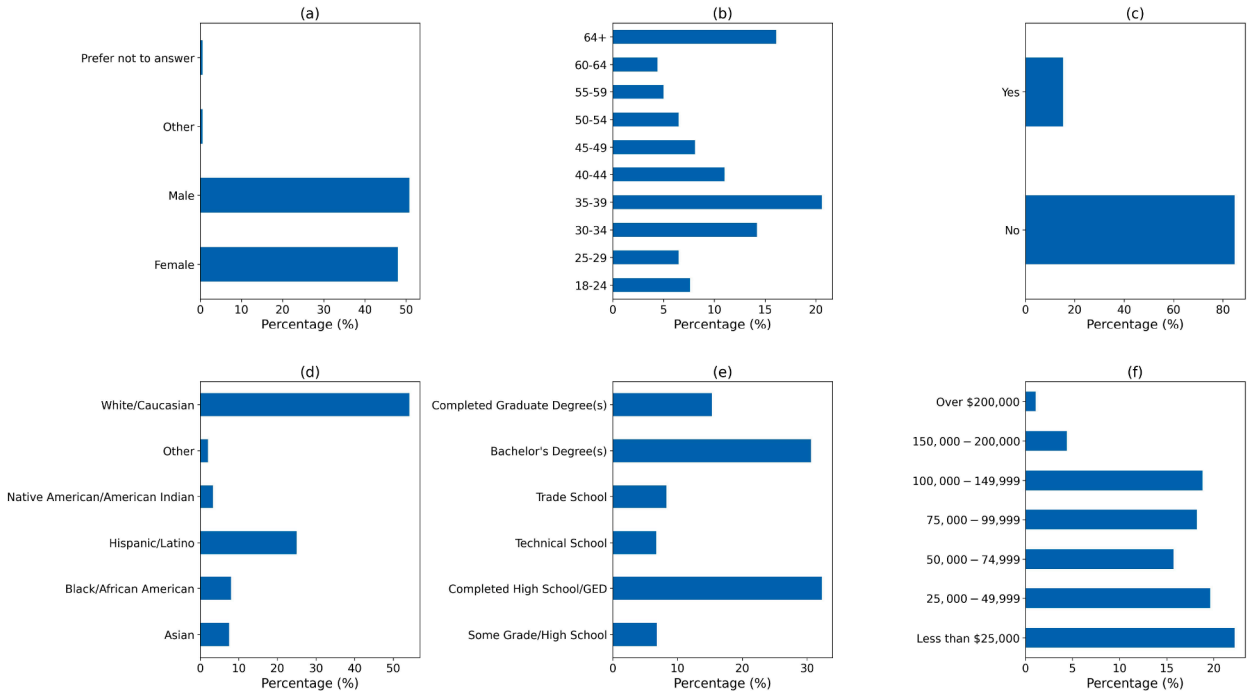


Fig. 3. Demographic profile of survey respondents, showing distributions by (a) gender, (b) age, (c) birthplace (born outside of the United States), (d) race/ethnicity, (e) highest level of education attained, and (f) household income.

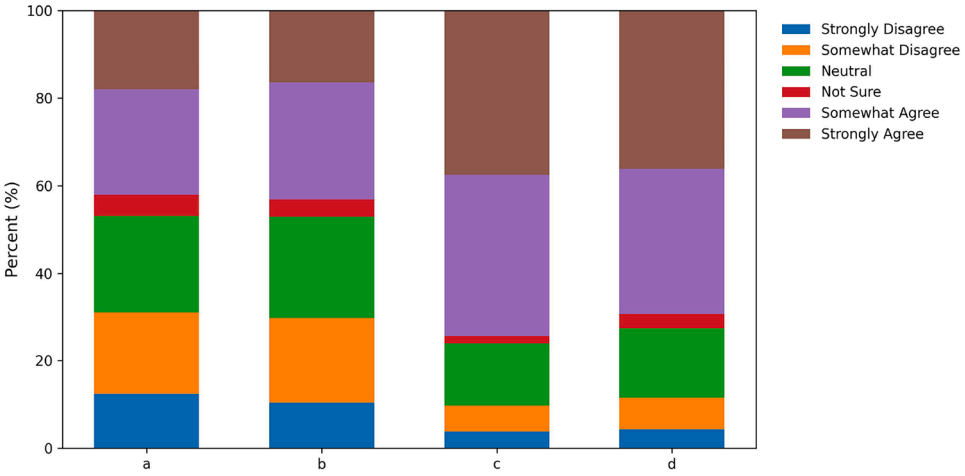


Fig. 4. Public attitudes toward expected outcomes of SSCs, showing respondent agreement with statements that SSCs: (a) make speeding worse outside camera zones; (b) increase travel time or congestion; (c) save lives and reduce injuries; and (d) make walking and cycling safer.

not view revenue collection as the primary purpose of these systems. Instead, the results indicate that public acceptance is closely tied to directing revenues toward visible, safety-oriented transportation benefits.

The revenue-related responses are consistent with long-standing international findings on SSC legitimacy. Prior studies repeatedly identify the perception that cameras are primarily revenue-raising as a major source of opposition, while also showing that acceptance improves when agencies demonstrate that camera siting and revenue allocation are grounded in safety goals rather than fiscal motives (Soole et al., 2008; Fleiter and Watson, 2012; Beaton et al., 2022). In this regard, the Oregon responses are not anomalous; rather, they reflect a broader pattern in which public support is strengthened when revenues are visibly reinvested into transportation safety improvements and when agencies avoid framing SSCs as a revenue tool.

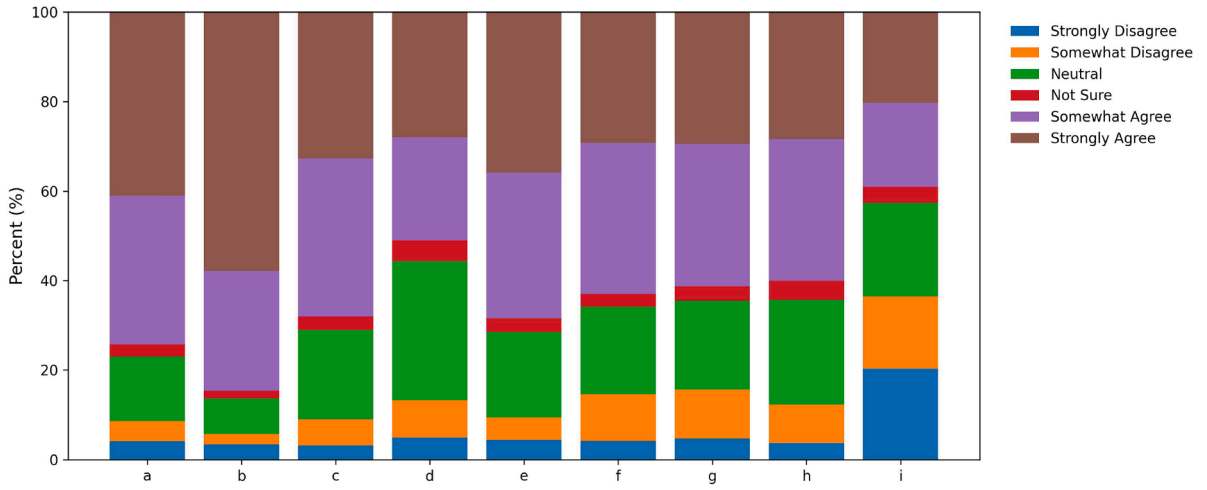


Fig. 5. Survey results on SSC location guidance, showing public attitudes toward: (a) construction zones; (b) school zones; (c) community-nominated locations; (d) locations intended to avoid disproportionate impacts; (e) locations where police cannot operate safely; (f) local low-speed city roads; (g) higher-speed state roads; (h) locations based solely on crash and speed statistics; and (i) opposition to Speed Safety Camera deployment on Oregon roads.

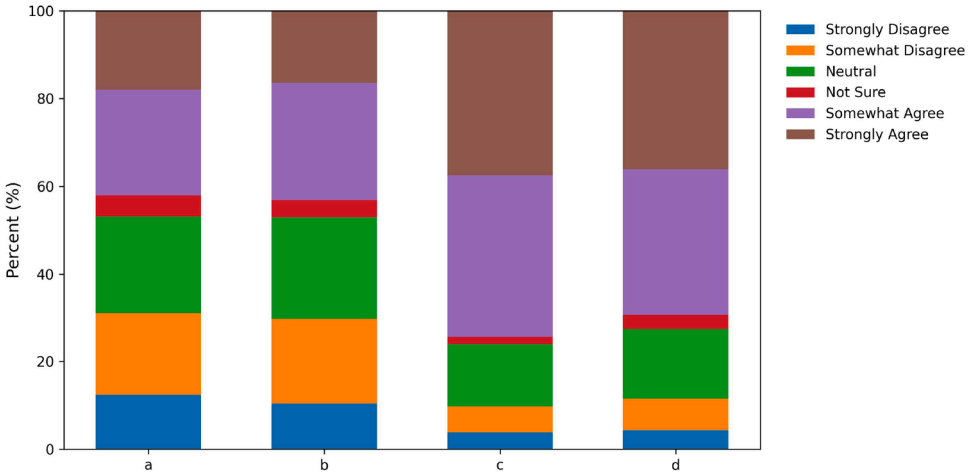


Fig. 6. Public attitudes toward existing conditions and beliefs related to SSCs, showing respondent agreement with statements that: (a) speeding is a problem in the respondent's neighborhood; (b) speeding is a problem in the city where the respondent lives or works; (c) SSCs invade privacy even without facial verification; and (d) mobile or hidden cameras lower speeds everywhere.

4.2. System architecture

This study implements a community-driven decision-support system for SSC policymaking by integrating LLMs, structured knowledge graphs, and unstructured document retrieval in a unified RAG framework. This architecture operationalizes the formal procedure defined in Section 3.4. The system is designed to generate transparent, context-aware, and actionable outputs by leveraging a hybrid reasoning pipeline that combines semantic search, graph-based inference, and natural language generation. To ensure reliability and interpretability, the system is strictly constrained to generate responses based only on retrieved context, thereby minimizing unsupported or speculative outputs. It should be noted that the retrieval process is relevance-based and does not impose explicit source-level weighting; instead, prioritization between authoritative and community-derived information is handled during the LLM reasoning stage and through human-centered interpretation.

The entire architecture is powered by a large language model operating in a zero-shot, prompt-based configuration without fine-tuning, which enables flexible and domain-adaptable generation grounded in retrieved context. Notably, the proposed framework is model-agnostic and not tied to any specific LLM; while a commercial model (i.e., GPT-4o) was used for prototyping and benchmarking, the system is designed to support a wide range of alternatives, including open-source and locally deployable models. This flexibility allows public agencies to adopt the framework in a manner consistent with governance, transparency, and data sovereignty

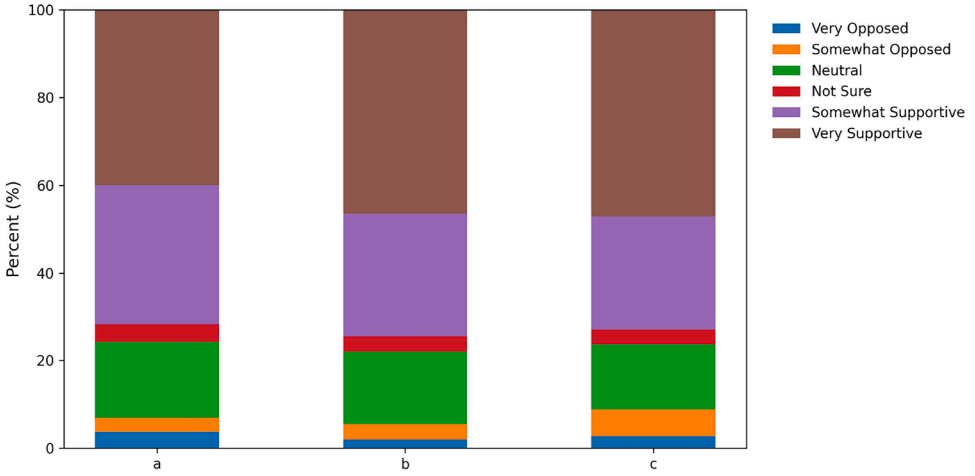


Fig. 7. Public attitudes toward transparency and accountability measures for SSCs, showing respondent agreement with: (a) reporting program results, including speeds, crashes, and injuries; (b) requiring disclosure of revenue use and citation information; and (c) making camera locations publicly known.

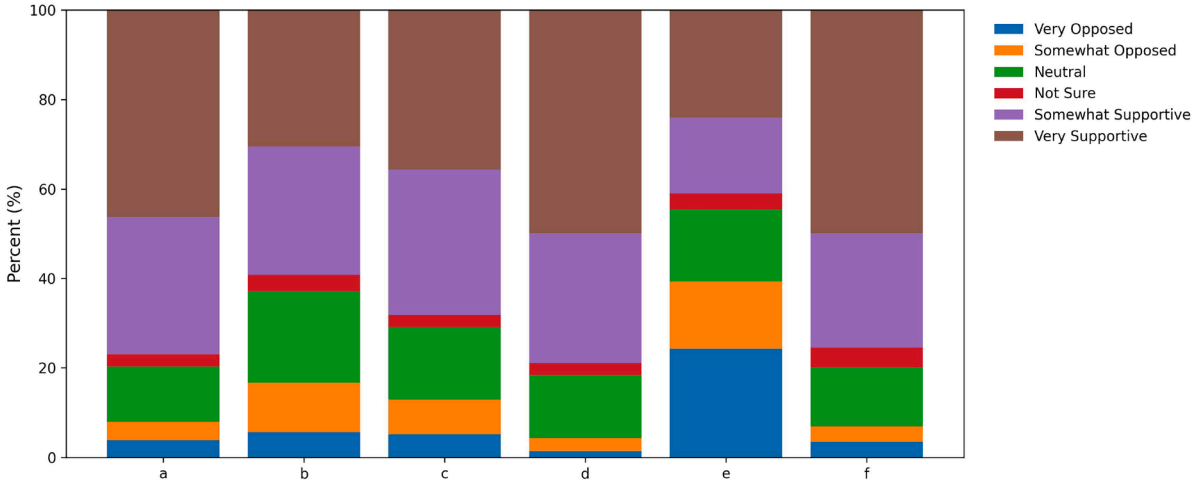


Fig. 8. Public attitudes toward SSC ticketing policy design, showing respondent agreement with: (a) applying different speed thresholds in school or work zones; (b) maintaining the same speed threshold but imposing higher fines in school or work zones; (c) requiring repeat offenders to pay higher fines; (d) allowing an educational course to forgive a first citation; (e) using income-based ticket pricing; and (f) issuing citations to drivers rather than vehicle owners.

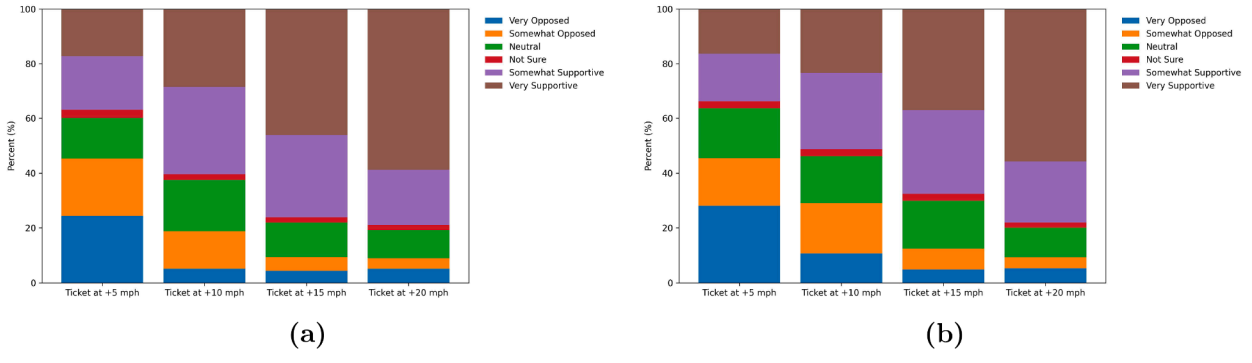


Fig. 9. Public preferences for SSC ticketing thresholds, showing attitudes toward issuing citations at 5, 10, 15, and 20 mph above the posted speed limit on (a) low-speed local roads and (b) high-speed state roads.

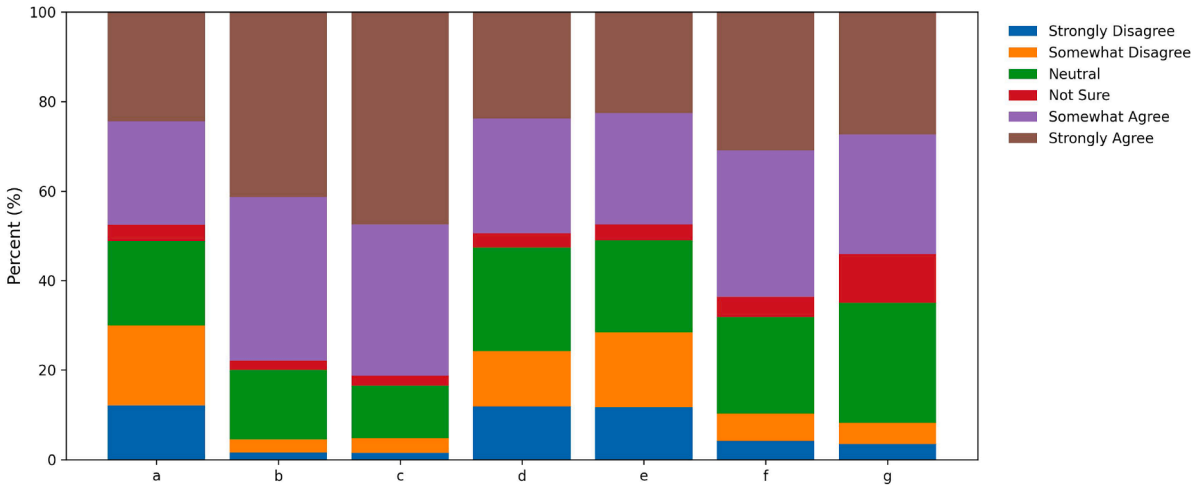


Fig. 10. Public views on SSC revenue use, showing respondent attitudes toward: (a) the perception that SSCs are primarily revenue-generating rather than safety-oriented; (b) using SSC revenues to improve safety on high-speed or high-crash roads; (c) allocating revenues to roadway infrastructure repair; (d) directing revenues to policing; (e) funding non-transport uses such as parks or schools; (f) using revenues to improve walking and cycling safety; and (g) the belief that SSC programs generate more revenue than their operational costs.

requirements. Furthermore, the framework follows a human-centered decision-support paradigm in which policymakers and domain experts interpret the generated outputs. In this way, the system is intended to augment expert decision-making by organizing and synthesizing complex information rather than replacing human judgment or creativity.

The proposed architecture is designed to be model-agnostic and future-compatible. Key components such as chunk size, retrieval depth, and prompt length are not fixed, but instead configurable based on the capabilities of the underlying language model. As modern LLMs support longer context windows, the system can incorporate larger inputs and advanced retrieval strategies to improve recommendation completeness and fidelity.

4.2.1. Knowledge base construction

The foundation of the system lies in constructing a comprehensive knowledge base that captures diverse sources of information relevant to SSC deployment. This includes authoritative documents comprising three key components. First, FHWA's Speed Enforcement Camera Systems Operational Guidelines provides comprehensive national guidance on planning, implementing, and evaluating SSC programs. It outlines best practices, legal considerations, enforcement operations, and program evaluation strategies, emphasizing the role of SSCs in reducing speeding-related crashes (Federal Highway Administration, 2008). Second, House Bill 4109, an Oregon legislative act, expands the legal framework for SSC deployment by streamlining photo radar enforcement laws and authorizing broader use by traffic enforcement agents (Oregon Legislative Assembly, 2024). Third, real-world case studies from jurisdictions including Portland, New York City, Philadelphia, Seattle, and the District of Columbia, as well as international examples from France and New South Wales, offer practical insights into SSC program design, equity considerations, public outreach, and effectiveness in reducing speeding and crashes (Figliozzi et al., 2025). These authoritative documents were consolidated with the full textual analysis of public survey responses, a summary of which was presented in the Survey Results section. The purpose of this integration is twofold: (1) to reflect real-world policy context, and (2) to amplify community-driven insights by embedding them into a scalable, machine-readable format.

The comprehensive knowledge base was then decomposed into semantically meaningful chunks using a configurable chunking strategy Eq. (5). In this implementation, a chunk size of 512 tokens (i.e., text units like words or subwords mapped to unique IDs for model processing) with an overlap of 24 was selected to facilitate both symbolic reasoning and retrieval-based processing, corresponding to $\tau = 512$ and $\omega = 24$ in Algorithm 1. This configuration is not a structural constraint of the framework; rather, chunk size and overlap remain tunable and can be adjusted as LLM context windows and embedding techniques evolve to better preserve semantic coherence and improve recommendation quality.

Recent studies provide empirical support for LangChain's experimental LLMGraphTransformer as an effective framework for extracting entities and relationships from unstructured text. Yamamoto et al. compared LLMGraphTransformer with other LLM-based knowledge graph construction methods and reported strong recall and competitive F1-scores, demonstrating its capability to capture semantically meaningful relationships from complex academic text (Yamamoto et al., 2025). Similarly, da Silva et al. showed that LangChain-based graph extraction approaches can be effectively customized for domain-specific applications, particularly for extracting structured nodes and relationships from user stories (da Silva, 2025). Although these studies noted that prompt engineering and ontology customization may improve performance in specialized domains, their findings collectively support the effectiveness of the LangChain graph-transformation framework for automated knowledge graph construction. Accordingly, we employed the LangChain's experimental LLMGraphTransformer (from the `langchain_experimental.graph_transformers.llm` python package)

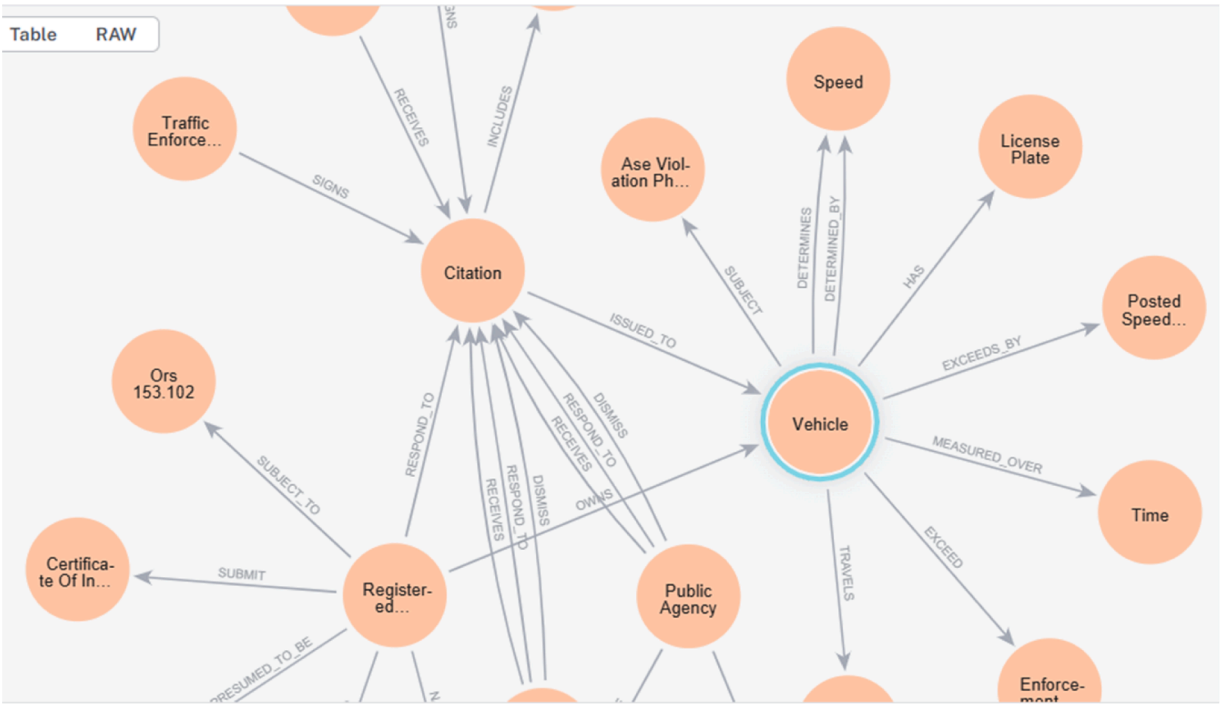


Fig. 11. A sample of the knowledge graph.

(LangChain Developers, 2024) to extract entities (e.g., “fine structure,” “mobile enforcement,” “equity evaluation,” “school zones,” “dangerous driving programs”) and their relationships (e.g., “mandated by,” “influences,” “recommended in,” “uses,” “covers,” “feedback from”), forming a domain-specific knowledge graph, i.e., the $\Phi_{\text{LLM}}^{\text{KG}}$ extraction step of Eq. (6). Each entity was enriched with descriptive properties such as corridor IDs, penalty levels, and program characteristics, which enables traceability of the recommendations generated downstream. The resulting graph comprises 3175 labeled nodes representing concepts, agencies, procedures, and public concerns, and 7321 labeled edges capturing their interdependencies. The entire graph is stored and queried in a Neo4j graph database, allowing for semantic traversal and interpretability. This structured graph serves as the semantic backbone of the recommender system, enabling reasoning about causal, procedural, and legislative linkages in SSC policy.

According to Fig. 11, we have shown a small sample of our knowledge graph. The knowledge graph illustrates the multi-faceted connections among key entities in a SSC system, with the Vehicle node at the center of a dense network of relationships. As shown in Fig. 11, a vehicle is linked to attributes such as Speed, License Plate, and Posted Speed Limit through relationships like DETERMINED_BY, HAS, and EXCEEDS_BY, respectively, and serves as the basis for determining whether a Citation is issued. The Citation itself is connected to multiple nodes, including the Registered Owner, Public Agency, Traffic Enforcement Officer, and SSC Violation Photo, which reflects the legal, administrative, and evidentiary processes involved in automated enforcement. These entities are further connected to statutory elements like ORS 153.102 and procedural components like the Certificate of Innocence, enabling dismissal or dispute resolution. This web of interactions highlights the integrated legal, operational, and data-driven components of the SSC system, emphasizing how each node can simultaneously participate in multiple relationships that collectively govern citation issuance, response, and enforcement.

4.2.2. Semantic indexing and hybrid retrieval

To complement the structured graph with unstructured semantic insights, we applied a dual retrieval strategy. On one side, the knowledge graph supports symbolic traversal for structured query responses, leveraging explicit relationships between entities. On the other side, a dense vector index was constructed by embedding the policy documents into a high-dimensional semantic space using the text-embedding-3-small model, forming the vector index $\mathcal{V}$ of Eq. (7). Documents and survey reports were chunked into semantically coherent segments and embedded using this model; all embeddings were indexed using a similarity search engine to support fast vector-based retrieval.

When a user submits a policy query, the system performs two retrieval operations in parallel, as formalized in Algorithm 2. First, user questions are parsed to extract entities using an LLM prompt-based schema, as shown in Table 1. A full-text fuzzy search is then performed in Cypher, followed by a bi-directional traversal of relationships for each matched entity to gather structured graph context Eq. (9). Second, it performs semantic similarity search over the embedded document corpus to retrieve the most contextually relevant text passages. For each query, the top four most similar chunks, i.e., $n = 4$ in Eq. (10), are retrieved using a hybrid search that combines vector search (semantic search), which relies on cosine similarity, and keyword search (sparse search), which uses exact

Table 1
LLM prompt-based schema for extracting entities.

Task	Template for Prompts
Defining the "question"	You are a transportation policy advisor. Given the topic of Speed Safety Cameras (SSCs) and relevant legislative, safety, and equity concerns, generate a structured set of policy and implementation recommendations. Please provide detailed content under the following categories: General, Introduction, Planning (Strategic, Program, and Equity), Enforcement and Field Operations, Administration and Violation Structure, Program Start-Up, Evaluation, Resources, Equity Considerations, Fine Structure, Covert Enforcement, Dangerous Driving Mitigation, Recommendations Based on Public Survey Results (including: Safety and Outcomes, Location Guidance, Public Beliefs, Transparency, Ticketing, and Revenue Allocation). Each category should include actionable guidance and may optionally highlight additional research opportunities or pilot programs. Write in a clear, policy-oriented tone as if preparing a guidance document for ODOT.
Extracting any entities inside the question	("system", "You are extracting all entities from the text.",), ("human", "Use the given format to extract information from the following input: {question}",)

Table 2
The custom chat prompt template.

Task	Template for Prompts
Keeping the chat history	Given the following conversation and a follow up question, rephrase the follow up question to be a standalone question, in its original language. Chat History: {chat_history} Follow Up Input: {question} Standalone question:
Defining the "context"	Structured data: {structured_data} Unstructured data: {"#Document". join(unstructured_data)}
Final unified prompt	Answer the question based only on the following context: {context} Question: {question} Please use natural language, and refrain from using knowledge beyond the above context. Answer:

term matching. These two sources of information, which include structured graph-based reasoning and unstructured document-based retrieval, are then merged to provide a comprehensive, multimodal context for the language model Eq. (11). This hybrid approach ensures that the system benefits from both explicit domain knowledge and latent textual patterns that may not be captured by symbolic structures alone.

4.2.3. Question understanding and prompt composition

To enable flexible interaction, the system accommodates both standalone and follow-up questions. For multi-turn conversations, follow-up queries are automatically reformulated into standalone questions using an LLM-based rephrasing module shown in Table 2, which condenses dialogue history into a context-independent form Eq. (8). This design is intended to improve conversational coherence and support more natural user engagement, although it does not guarantee that every requirement introduced in earlier turns will be retained. Query reformulation and repeated retrieval can reduce, but not eliminate, context loss. Therefore, evaluating and improving long-horizon multi-turn reliability remains an important direction for future research (Laban et al., 2024, 2025).

After retrieval, the system synthesizes the structured (graph-based) and unstructured (document-based) information into a unified prompt format, the structure of which is defined by a custom chat prompt template Eq. (12), as presented in Table 2. The merged context includes graph-derived insights such as "`Hb4109 - AFFECTS -> Ssc Program`," alongside text passages from documents offering detailed explanations. In addition, the system incorporates query embeddings and multi-turn chat history (if applicable) into the prompt. This prompt is then passed to an LLM, operating at `temperature = 0` to ensure deterministic and grounded outputs. Also, we set `max_tokens = 2048` to limit the model's output length, ensuring responses are concise, relevant, and computationally efficient. The prompt is deliberately constrained to discourage hallucinations and ensure the model's response remains grounded in retrieved evidence. This structured prompt engineering approach allows LLM to conduct domain-specific reasoning without needing fine-tuning.

The final output consists of structured policy recommendations organized under predefined categories provided in the prompt. These are grouped into two primary categories and 18 subcategories, as illustrated in Fig. 12. Each category is populated with actionable guidance derived from the fused context, often supplemented with pilot suggestions, equity-focused considerations, or data-driven thresholds. Table 3 presents a comparison between recommendations generated by the LLM and those manually provided by domain experts (Figliozzi et al., 2025) across two sample subcategories. In addition, comprehensive comparisons of LLM-generated and expert recommendations across all policy subcategories within the "SSC Guidance Additions" and "Recommendations From

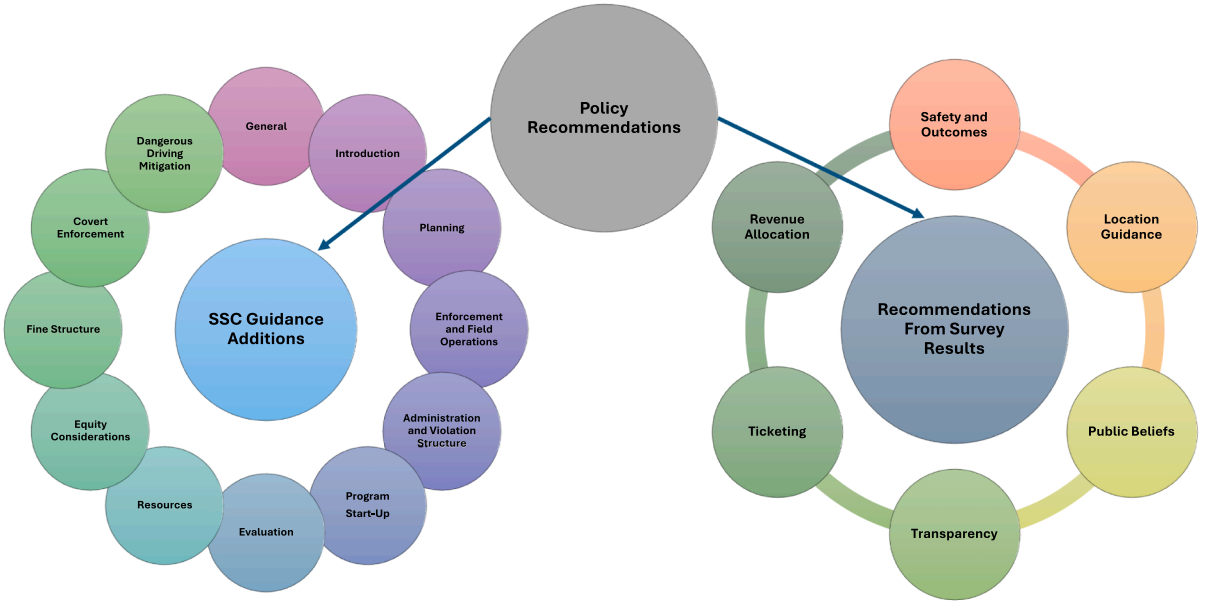


Fig. 12. Categorization of policy recommendations.

Table 3
Comparison of LLM-generated and expert recommendations by policy subcategory.

Subcategory	Expert-Provided Recommendation	LLM-Generated Recommendation
Fine Structure	Oregon could consider developing a more tailored fine structure with fine amounts specifically set for speed safety cameras. Fines can vary by speed, number of offenses, or income. Many jurisdictions with widespread SSC systems, such as Chicago or New York City, have a flat fee for violations. Additional analysis should be conducted to determine the fine amount and scale of the fine. One option is to have a fine that varies by speed or number of violations, and then have an option for a reduced fine based on income. Changes to the fine structure would involve amending ORS 810.439 to explicitly allow fines that differ from those issued by police officers for speeding. This fine could range from $50 to $100 for an initial violation and be escalated for subsequent violations or excessive speeds. An additional research opportunity for ODOT is to conduct research to develop a new fine structure, design a pilot study, and evaluate the pilot.	Graduated Fines: Implement a graduated fine system based on income or ability to pay. Provide alternative options for those unable to pay fines, such as community service or traffic safety classes.
Revenue Allocation	There is strong agreement that SSC-generated revenue should fund infrastructure repairs, safety improvements on high-speed and high-crash corridors, and law enforcement and micromobility enhancements. There are mixed opinions on whether SSC systems are primarily for safety improvements or revenue generation.	Allocate SSC revenue to safety projects and community improvements, particularly in underserved areas. Communicate revenue use to the public to build trust and support.

Survey Results" categories are reported in Tables A.5 and A.6, respectively, in the Appendix. These results show that while the LLM's recommendations generally align with those of experts, they are more concise, partly due to the maximum token limit set at 2048. However, thanks to the multi-turn conversational capability of our proposed approach, users can submit follow-up questions to elicit more detailed responses from the LLM-based system.

4.3. Evaluation

We evaluated system performance using two complementary approaches: (1) Sentence-BERT cosine similarity against expert ground truth, supported by a 10,000-iteration Monte Carlo simulation with kernel density estimation (KDE) visualized through violin plots to assess variability; and (2) RAGAS metrics (faithfulness, answer relevancy, context precision/recall, and context entity recall) to assess alignment among queries, retrieved context, and generated responses.

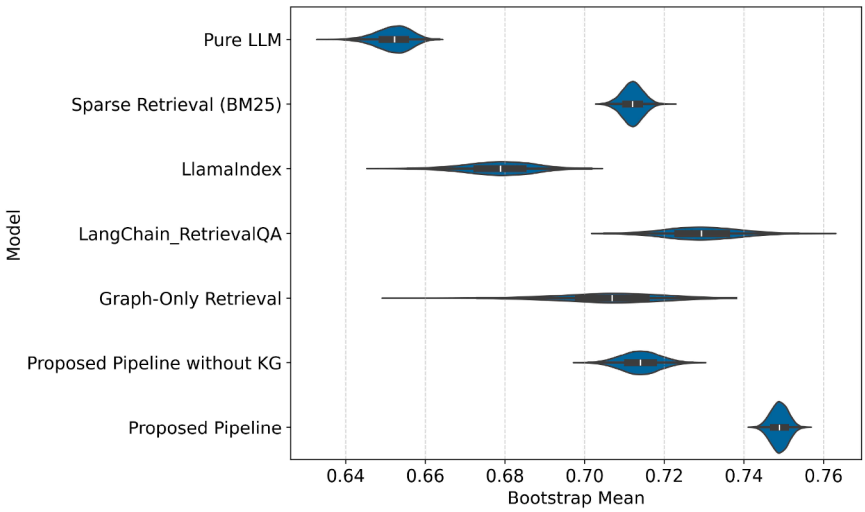


Fig. 13. Violin plot of bootstrap mean similarity scores for different approaches.

4.3.1. Similarity-based benchmarking

In the first strategy, we produced 10 model-generated outputs and benchmarked each against a manually curated ground-truth policy document using cosine similarity computed with Sentence-BERT (all-MiniLM-L6-v2 model) (Reimers and Gurevych, 2019). The ground truth for evaluation was provided manually by domain experts (Figliozzi et al., 2025). The evaluation was across 10 runs because LLM outputs are inherently non-deterministic, even when the temperature is set to 0, as was the case with LLM in our proposed approach. Generating multiple outputs allowed us to capture the variability in LLM responses and ensure a fair and robust evaluation.

To assess the reliability and variability of the similarity scores, we conducted a Monte Carlo bootstrap simulation (Harrison, 2010) with 10,000 iterations. In each iteration, samples were drawn with replacement from the set of similarity scores, allowing us to approximate the underlying distribution. This approach enabled us to estimate the statistical properties of the model's performance, such as mean, variance, and distribution shape. To further analyze the uncertainty, we applied KDE (Chen, 2017) to estimate the underlying probability density distributions, which were visualized through violin plots to provide a clearer understanding of score dispersion and the robustness of each method.

We compared our proposed pipeline against several widely used RAG baselines, including LangChain's RetrievalQA, LlamaIndex's RAG pipeline, sparse retrieval (BM25), graph-only retrieval, and a version of our approach without knowledge graphs. A consistent LLM backbone was used across all pipelines to ensure fair comparison. Its inclusion as a standalone baseline is also justified by its central role in our system and its prevalence in current LLM applications. LangChain's RetrievalQA is a flexible framework that helps create question-answering systems by combining language models with document retrieval. It works by searching through an extensive collection of documents to find relevant information and then using an LLM to generate a clear and accurate response (Sar et al., 2024). LlamaIndex's RAG pipeline is developed to connect LLMs with specialized datasets. It's especially helpful for creating tools like chatbots or search systems that need to pull in specific information quickly and respond with high-quality answers (Rothman, 2024). In addition, sparse retrieval (BM25) was included as a traditional lexical retrieval baseline due to its widespread adoption in keyword-based information retrieval (Thakur et al., 2021). In addition to external baseline comparisons, we conducted ablation-style evaluations by (1) removing the knowledge graph component from the proposed framework while preserving the same retrieval and generation backbone to assess the contribution of the knowledge graph component within the proposed pipeline, and (2) removing the vector-based semantic retrieval component while preserving the same structured knowledge graph traversal mechanism to evaluate the effectiveness of graph-based retrieval without semantic vector retrieval. These evaluations allow the individual contributions of structured graph reasoning and semantic retrieval to be assessed independently.

Fig. 13 presents the violin plot of bootstrap mean similarity scores for seven approaches, each evaluated over 10,000 iterations. The inner boxplots represent the distribution and central tendency of the mean similarity scores, while the violin shapes illustrate the density distribution of those means across simulations.

As shown in Fig. 13, among the approaches compared, the proposed hybrid KG and RAG pipeline outperforms all baselines, achieving the highest mean similarity score of 0.75, with a standard deviation of 0.0022 and a 95% confidence interval of (0.7446, 0.7531). In practical terms, a similarity score of 0.75 indicates that the generated output generally reflects the same key policy themes and recommendations as the expert guidance, although the wording and detail may differ. It should not be interpreted as 75% accuracy, but rather as a measure of overall meaning-level alignment with the expert guidance. Its narrow violin distribution indicates highly consistent performance across Monte Carlo simulations, reflecting both high accuracy and low variability. In contrast, the proposed pipeline without the knowledge graph component exhibits a lower mean similarity score and a wider violin distribution, indicating that the removal of structured knowledge integration reduces both performance and stability. The LangChain_RetrievalQA

approach also yields a lower similarity score (approximately 0.73) and shows greater variability, as evidenced by its broader violin distribution relative to the hybrid system. This suggests that, while retrieval augmentation provides measurable benefits, it does not achieve the same level of precision or consistency as the hybrid framework. The LlamaIndex approach performs noticeably worse than the hybrid pipeline, with a lower mean similarity score (approximately 0.68) and a violin distribution indicative of fluctuating performance across runs. The graph-only retrieval approach achieves moderate performance (approximately 0.71) but exhibits a wider distribution, suggesting higher variability due to the absence of vector-based semantic retrieval. Sparse retrieval (BM25) also performs worse than the proposed pipeline, with a mean similarity score around 0.71 and a relatively concentrated distribution, indicating that keyword-based retrieval alone cannot achieve the same level of contextual understanding and accuracy as the hybrid KG and RAG framework. Finally, the pure LLM baseline without retrieval augmentation yields the lowest similarity scores overall (approximately 0.65). Although its distribution is relatively narrow, its lower performance underscores the limitations of generation without retrieval or structured knowledge support. Taken together, these results demonstrate that the hybrid KG and RAG system not only achieves superior accuracy compared to the other methods but also exhibits greater robustness and stability, reinforcing the value of combining structured knowledge graphs with unstructured retrieval for producing context-aware, policy-relevant outputs.

To further contextualize the robustness of the evaluation results, we compare the observed similarity scores with those reported in previous works that utilize RAG-based architectures in safety-critical domains. Across four previous studies, the semantic similarity score typically ranges from approximately 0.68 to 0.88, spanning applications such as medical question answering (QA), construction safety, clinical reasoning, and telecom QA (Sivakumar et al., 2026; Lee et al., 2024; Potluri et al., 2025; Roychowdhury et al., 2024). The similarity score of 0.75 reported in this study falls well within this range, matching or exceeding several baseline RAG configurations and aligning with strong LLM-based models, although remaining slightly below the highest-performing fine-tuned systems. This difference is expected, as prior studies primarily address structured QA tasks with well-defined ground truth, whereas the proposed framework tackles a more complex policy recommendation problem using a heterogeneous RAG-based architecture. Overall, consistent with prior literature, these results indicate that the model effectively grounds recommendations for safety-critical applications, confirming that a similarity score of 0.75 is both competitive and robust.

4.3.2. RAGAS metrics

In the second assessment strategy, we assessed the performance of the system's RAG pipeline by examining how well the query, retrieved context, and generated answer aligned in terms of relevance. This evaluation was conducted using RAGAS (Es et al., 2024), an open-source framework specifically designed for measuring the effectiveness of RAG systems. This framework needs four key components from the pipeline: the user query, the model's output, the retrieved context, and the reference ground truth answer, all of which were already obtained during the modeling process. The assessment relies on a set of defined metrics (Es et al., 2024; Chen et al., 2024):

- Faithfulness: This metric measures how well a response's claims are supported by the retrieved context, with scores ranging from 0 to 1. A response is considered faithful when all claims can be inferred from the retrieved context. The faithfulness score is computed as:

$$\text{Faithfulness Score} = \frac{\text{Number of response claims supported by the retrieved context}}{\text{Number of claims in the response}} \tag{13}$$

- Answer Relevancy: This metric measures how well a response aligns with the intent of the user input, with higher scores indicating better relevance. It is computed by averaging the cosine similarity between the user input embedding and embeddings of questions generated from the response:

$$\text{Answer Relevancy} = \frac{1}{N}\sum_{i=1}^{N}\frac{E_{g_i}\cdot E_o}{\|E_{g_i}\|\|E_o\|} \tag{14}$$

  where:
  - E_{g_i} is the embedding of the i-th generated question,
  - E_o is the embedding of the user input,
  - N is the number of generated questions (default $N = 3$).
- Context Precision: It evaluates how well relevant context chunks are ranked above irrelevant ones in the retrieved results. It is calculated by averaging precision@k values across the ranked context. The metric is computed as:

$$\text{Context Precision@}K = \frac{\sum_{k=1}^{K}\left(\text{Precision@}k \times v_k\right)}{\text{Number of relevant items in the top } K} \tag{15}$$

$$\text{Precision@}k = \frac{\text{true positives@}k}{\text{true positives@}k + \text{false positives@}k} \tag{16}$$

  where:
  - K is the total number of chunks in the retrieved context,
  - $v_k \in \{0, 1\}$ indicates whether the chunk at rank k is relevant.

Table 4
Evaluation metrics for different RAG pipelines.

RAG Pipeline	Faithfulness	Answer Relevancy	Context Precision	Context Recall	Context Entity Recall
Sparse Retrieval (BM25)	0.63	0.70	0.92	0.13	0.06
LlamaIndex	0.61	0.69	1.00	0.20	0.10
LangChain_RetrievalQA	0.79	**0.72**	0.71	0.11	0.07
Graph-Only Retrieval	0.58	0.71	0.92	**0.48**	0.18
Proposed Pipeline without KG	0.69	0.70	1.00	0.15	0.05
Proposed Pipeline	**0.87**	0.70	**1.00**	0.26	**0.20**

- Context Recall: It measures how much of the relevant information is captured by the retrieved context. It is computed by comparing claims in a ground truth answer against the retrieved context and assessing how many are supported. The score is defined as:

$$\text{Context Recall} = \frac{\text{Number of claims in the ground truth supported by the retrieved context}}{\text{Number of claims in the ground truth}} \tag{17}$$

- Context Entity Recall: It measures the proportion of ground truth entities that are present in the retrieved context, reflecting how well entity-level information is recovered. The metric is defined as (Es et al., 2024; Chen et al., 2024):

$$\text{Context Entity Recall} = \frac{\text{Number of common entities between } RCE \text{ and } RE}{\text{Number of entities in } RE} \tag{18}$$

  where:
  - RE is the set of entities in the ground truth,
  - RCE is the set of entities in the retrieved contexts.

The evaluation results across the baseline pipelines are presented in Table 4.

According to Table 4, the evaluation results show that the proposed Hybrid KG and RAG pipeline outperforms all baseline RAG approaches across most metrics. It achieves the highest faithfulness score (0.87) and the strongest context entity recall (0.20), indicating superior factual grounding and more comprehensive retrieval of relevant entities and information. The proposed pipeline also maintains competitive answer relevancy (0.70), strong context recall (0.26), and perfect context precision, demonstrating reliable retrieval quality without introducing irrelevant context. In comparison, LlamaIndex exhibits lower faithfulness (0.61) and moderate answer relevancy (0.69), coupled with substantially weaker context recall (0.20), suggesting limited coverage of relevant information despite accurate ranking of retrieved chunks. LangChain_RetrievalQA achieves relatively high faithfulness (0.79) and the highest answer relevancy (0.72), but performs poorly in context recall (0.11) and context entity recall (0.07), indicating that it often fails to retrieve sufficient supporting information even when generated answers appear relevant. Sparse Retrieval (BM25) exhibits low faithfulness (0.63) and very weak context recall (0.13) and context entity recall (0.06), suggesting that keyword-based retrieval alone struggles to capture semantically relevant supporting information. The graph-only retrieval approach achieves the highest context recall (0.48) among all methods but shows lower faithfulness (0.58), suggesting that structured graph traversal alone retrieves broader contextual information while lacking the semantic grounding provided by the hybrid framework. The proposed pipeline without the knowledge graph component shows intermediate performance, highlighting the contribution of structured knowledge integration. Overall, these results demonstrate that combining structured knowledge graphs with semantic retrieval enables the LLM to access both symbolic and unstructured context, improving factual grounding, reducing hallucinations, and increasing retrieval completeness. Consequently, the Hybrid KG and RAG pipeline delivers more balanced and robust performance than approaches relying on a single retrieval strategy.

To further contextualize the robustness of the evaluation results, we benchmark the observed RAGAS metrics against prior studies employing RAG-based architectures in safety-critical domains. The robustness of the RAGAS results in this study is broadly consistent with prior RAG-based systems in safety-critical domains, which report strong faithfulness (≈0.57-0.94), answer relevancy (≈0.71-0.91), and, in some cases, high context recall or near-perfect precision using models such as GPT-4o, GPT-3.5, Mistral-7B, and LLaMA-based pipelines (Ammar et al., 2025; Roychowdhury et al., 2024; Chen et al., 2024; Baur et al., 2025; Pires et al., 2025). Compared to these studies, the proposed framework achieves similarly strong or higher performance in faithfulness (0.87) and context precision (1.00), indicating reliable grounding and highly accurate retrieval, while answer relevancy (0.70) remains competitive but slightly lower due to the open-ended nature of policy recommendation. However, context recall (0.26) and context entity recall (0.20) are lower, which is expected because prior work operates in closed, well-structured domains with aligned QA tasks, whereas this study addresses heterogeneous, multi-source policy reasoning where selective retrieval and interpretability are prioritized over exhaustive coverage. Overall, the results align well with prior safety-critical RAG systems in terms of grounding and precision, with differences in recall reflecting the broader and more complex nature of the policy recommendation task rather than limitations of the approach.

5. Discussion and ethical considerations

A limitation of the current study is that the empirical validation is based on a single-state case (Oregon). While the system demonstrates strong performance within this context, its broader applicability depends on the availability and integration of jurisdiction-specific data sources. Future work should include multi-state validation to further assess generalizability. To support

cross-jurisdictional adoption, we outline a process-driven implementation perspective inspired by recent applications of generative AI in risk management, where AI systems are embedded within continuous decision-support processes (Leonard et al., 2026). The proposed SSC recommender system can be adapted through several interrelated steps. First, Oregon-specific inputs, including legislation, survey data, and institutional context, can be replaced with jurisdiction-specific data sources to ensure local relevance. Second, the knowledge graph can be reconstructed using locally defined policy entities while preserving the underlying ontology. Third, retrieval and reasoning components can be recalibrated using region-specific documents, such as state DOT guidance and municipal policies, to enable context-aware recommendation generation; concretely, this recalibration is carried out through the free parameters of the formal procedure in Section 3.4 (τ, ω, n, and δ in Eqs. (5), (10), and (9)), which can be re-tuned per jurisdiction without altering the underlying architecture. Finally, the system should be continuously updated with new data and policy changes while maintaining human-centered interpretation and oversight. This process-oriented design helps maintain adaptability across diverse institutional and regulatory contexts while preserving methodological rigor.

The deployment of AI-driven tools in transportation policy requires careful consideration of governance, accountability, and system limitations. Unlike commercial recommender systems, policy guidance systems influence regulatory and societal outcomes, necessitating a human-centered decision-support paradigm. Accordingly, the proposed framework is designed to support, rather than replace, decision-making by enabling policymakers and domain experts to interpret its outputs, while retaining full responsibility for implementation. The system enhances traceability through knowledge graphs and retrieval-based grounding, allowing users to inspect the evidence underlying each generated output. However, as a retrieval-augmented framework, it remains constrained by the scope and quality of its data sources and primarily synthesizes existing knowledge rather than generating novel policy concepts. Thus, it should be viewed as a tool for strengthening evidence-based reasoning rather than enabling fully autonomous innovation.

The survey was reviewed by the Oregon State University Human Research Protections Program (HRPP) and determined to be exempt from Institutional Review Board (IRB) review (Protocol No. HE-2024-1207). The exemption falls under Category 2(i), as the research involves anonymous survey procedures in which no identifiable personal information was collected. There are potential survey-related biases. Online panel-based surveys may underrepresent populations without internet access or those not enrolled in survey panels, and certain socio-demographic groups may still be under- or over-represented despite the use of quota controls. These challenges are partially mitigated through post-stratification weighting; however, they cannot be fully eliminated. Future work should explore hybrid data collection approaches, including in-person and community-based sampling, to better capture underserved populations and further reduce bias.

The use of LLM-based recommender systems in policy contexts introduces risks of algorithmic bias, including model-induced bias. To address these concerns, this study implements multi-level mitigation strategies. At the knowledge level, heterogeneous sources, including survey data, legislation, and national guidance, are integrated to reduce reliance on single perspectives. At the model level, RAG and knowledge graphs constrain outputs to contextually grounded evidence. Another limitation of the current framework is the absence of explicit hierarchical weighting among information sources. In policy contexts, regulatory requirements should take precedence over public sentiment. Future work should incorporate constraint-aware retrieval and ranking mechanisms that explicitly encode source hierarchy, such as legislation as hard constraints and public opinion as soft signals, to improve reliability for real-world policy deployment.

For real-world public-sector deployment, reliance on proprietary models may introduce risks related to cost and availability. To address this, future implementations should prioritize open-source or publicly governed LLMs to improve reproducibility and institutional control. Another limitation is the rapid evolution and potential deprecation of specific LLMs, which may create operational vulnerabilities. However, the modular design of the proposed framework mitigates this risk by enabling seamless substitution of the underlying LLM without altering the overall architecture. The observed differences between expert-provided and LLM-generated recommendations, particularly in contextual richness, may partly result from limitations in prompt length. While the framework retrieves relevant context effectively, the amount of information passed to the LLM may limit response depth. As LLM context windows expand, future implementations can incorporate larger contextual inputs and more advanced retrieval strategies to improve recommendation completeness. Although the proposed framework outperformed conventional RAG baselines, future work should compare it against emerging agentic and multi-agent retrieval architectures to further evaluate scalability and robustness.

Despite its promising performance, LangChain's experimental LLMGraphTransformer framework may still introduce occasional semantic inaccuracies when extracting entities and relationships from complex policy documents. Although expert-assisted validation of the extracted entities and relationships was not conducted in this study due to time and resource limitations, it remains an important direction for future research to improve extraction reliability.

6. Conclusion

This study introduces a novel, community-informed policy recommender system that integrates LLMs, KGs, and RAG to support transparent and evidence-based SSC deployment in Oregon. By embedding public sentiment from a statewide survey and authoritative documents into a unified RAG-KG architecture, the system addresses persistent concerns of fairness, legitimacy, and policy clarity that often undermine SSC programs. Experimental results, which were validated through Monte Carlo simulations, sentence similarity benchmarks, and RAGAS metrics, demonstrate that the proposed hybrid approach significantly outperforms traditional LLM and RAG pipelines in faithfulness, contextual grounding, and interpretability. Beyond technical superiority, the framework's scalability offers an adaptable and process-driven approach for transportation agencies seeking to harmonize data-driven safety interventions with community trust and policy compliance. This co-production of AI-driven insights and public values marks a meaningful advancement in equitable, explainable, and future-ready transportation governance.

Future work will focus on expanding the system's generalizability by incorporating multi-state or national survey datasets and conducting more extensive data collection across diverse jurisdictions. Enhancements to the knowledge graph will include multi-modal integration, expressive graph structures, and scalable enrichment to improve knowledge quality. Additionally, integrating user feedback loops into the recommendation process will further enhance interpretability and adaptability for practical policy decision-making.

CRediT authorship contribution statement

Farhad Sedighi: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization; **Brian M. Staes:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Investigation, Formal analysis, Conceptualization; **Benyamin Ghoreishi:** Writing – review & editing, Investigation, Formal analysis, Conceptualization; **Xuehan Jing:** Writing – review & editing, Investigation, Conceptualization; **Haizhong Wang:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Investigation, Funding acquisition, Conceptualization.

Declaration of competitng interest

The authors declare no conflict of interest related to the content or findings of this study.

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Data availability

Data will be made available on request.

Appendix A.

Table A.5
Comparison of LLM-generated and expert recommendations across policy subcategories in the "SSC Guidance Additions" category.

Subcategory	Expert-Provided Recommendation	LLM-Generated Recommendation
General and Introduction	Expand the scope of ODOT's guidance to include both fixed and mobile SSC on state highways, statewide. The current scope focuses only on fixed SSC on state highways. Develop guidance and recommendations for non-state facilities. Consider setting a recommended limit on the number of SSC devices based on population. Update using the latest relevant legislation.	Speed Safety Cameras (SSCs) are a critical component of modern traffic management systems, designed to enhance road safety by monitoring and enforcing speed limits. This document outlines a comprehensive set of policy and implementation recommendations for the Oregon Department of Transportation (ODOT) to effectively deploy and manage SSCs, addressing legislative, safety, and equity concerns.
Planning	Strategic Planning should include a safety needs assessment, legal and policy review, identification of stakeholders, and an expansion of Safety and Operations reporting. Program Planning should set goals for the program, determine the scope, scale, and type of program, and develop an SSC program plan. Equity should be addressed by including guidance for equity in site selection, operations, and evaluation.	Strategic Planning: Develop a long-term vision for SSC deployment that aligns with the Safe System Approach and Vision Zero goals. Prioritize high-crash network streets and school zones for initial implementation. Program Planning: Establish clear selection criteria for SSC locations, focusing on areas with high rates of speed-related crashes. Utilize data-driven approaches to identify priority zones. Equity Planning: Conduct equity analyses to ensure SSC placement does not disproportionately impact marginalized communities. Revenue from SSCs should be reinvested in low-income areas and those with high injury rates, as informed by environmental health disparities maps.
Enforcement and Field Operations	Enforcement should provide guidance to jurisdictions on how to conduct enforcement operations. Field Operations should include a guidance or template for jurisdictions on establishing a Field Operations Plan, which would offer internal guidance on personnel and equipment logistics.	Enforcement Strategy: Implement a mix of fixed and mobile SSCs to cover a broad range of locations. Ensure that enforcement is consistent and transparent to maintain public trust. Field Operations: Collaborate with vendors like Verra Mobility for the installation and maintenance of SSCs. Ensure that data from SSCs is accessible and integrated into ODOT's traffic management systems.
Administration and Violation Structure	Outline the violation structure, including options for distribution of responsibilities, steps in a violation issuance process, options for citation recipients, status tracking and follow-up procedures, and handling of violation notices issued to government vehicles.	Violation Process: Develop a streamlined citation issuance process that includes warnings for first-time offenders. Work with local courts to establish income-based fines and alternative options for those unable to pay. Program Administration: Assign a dedicated program manager to oversee SSC operations, ensuring compliance with state laws and regulations.
Program Start-Up	Provide guidance on program start-up tasks including equipment procurement, vendor agreements, coordination with DMV, data management, public education, program roll-out, and a warning period.	Initial Deployment: Begin with a pilot program in select high-risk areas to evaluate the effectiveness of SSCs. Use findings to refine deployment strategies and expand the program. Public Outreach: Conduct active and passive outreach to educate the public about the benefits and operations of SSCs. Use community meetings, flyers, and online platforms to engage with residents.
Evaluation	Describe data collection for basic program monitoring, outline reporting requirements, and an evaluation procedure. Discuss the procedure for identifying dangerous drivers and strategies to change their behavior.	Performance Metrics: Measure the effectiveness of SSCs through reductions in speed-related crashes and repeat violations. Use public feedback and traffic diversion data to assess program success. Continuous Improvement: Regularly review and update SSC policies based on evaluation findings and emerging best practices.
Resources	Provide additional documentation and resources for jurisdictions.	Funding Allocation: Ensure that SSC revenue is allocated to traffic safety projects, particularly in low-income and high-crash areas. Explore additional funding sources to support program expansion. Vendor Partnerships: Maintain strong relationships with SSC vendors to ensure high-quality service and data provision.

Table A.5
Continue.

Subcategory	Expert-Provided Recommendation	LLM-Generated Recommendation
Equity	Oregon could study recommending or mandating a geographical distribution of cameras and violations to ensure no one community is unfairly targeted. Exploring a flat or income-dependent fine structure as cameras become more prominent could help reduce the financial burden on lower-income individuals. This could include general legislation mandating an equity analysis as part of site selection and/or legislation authorizing an income-based fine pilot in a specific city, such as Portland. An additional research opportunity is for ODOT to study and develop an equity analysis methodology specifically tailored for SSC location selection or evaluation and/or identify how existing ODOT demographic and equity tools can be applied for SSC programs.	Equity Analysis: Conduct thorough equity analyses for all new SSC locations. Ensure that camera placement and fine structures do not disproportionately affect vulnerable populations. Community Engagement: Involve community members in the planning process to address equity concerns and build public support.
Fines	Oregon could consider developing a more tailored fine structure with fine amounts specifically set for speed safety cameras. Fines can vary by speed, number of offenses, or income. Many jurisdictions with widespread SSC systems, such as Chicago or New York City, have a flat fee for violations. Additional analysis should be conducted to determine the fine amount and scale of the fine. One option is to have a fine that varies by speed or number of violations, and then have an option for a reduced fine based on income. Changes to the fine structure would involve amending ORS 810.439 to explicitly allow fines that differ from those issued by police officers for speeding. This fine could range from $50 to 100 for an initial violation and be escalated for subsequent violations or excessive speeds. An additional research opportunity for ODOT is to conduct research to develop a new fine structure, design a pilot study, and evaluate the pilot.	Graduated Fines: Implement a graduated fine system based on income or ability to pay. Provide alternative options for those unable to pay fines, such as community service or traffic safety classes.
Covert Enforcement	Oregon could consider piloting more covert, non-signed camera locations to investigate the effectiveness of these devices on decreasing speeds over a wider area or time. This would consist of legislation allowing a temporary pilot without requiring signage. The pilot could run for a specific period and undergo evaluation to determine its efficacy. An additional research opportunity for ODOT is to design and evaluate a covert enforcement pilot to understand the effectiveness of techniques such as no signage.	Halo Effect: Consider the use of covert SSCs to create a "halo effect," deterring speeding even in areas without visible enforcement.
Dangerous Driving	Oregon could consider piloting a dangerous driving program that impounds vehicles or installs intelligent speed assist on vehicles with excessive SSC violations. This would require more research on the relationship between SSC violators and crashes. An additional research opportunity for ODOT is to conduct studies linking SSC violations and crashes to understand whether individuals with more or higher-speed violations are more likely to be involved in injury crashes. This would inform the need for a program targeting dangerous drivers.	Targeted Programs: Develop programs to address dangerous driving behaviors, such as aggressive speeding and red-light running. Use SSC data to identify and target high-risk drivers.

Table A.6
Comparison of LLM-generated and expert recommendations across policy subcategories in the "Recommendations From Survey Results" category.

Subcategory	Expert-Provided Recommendation	LLM-Generated Recommendation
Safety and Expected Outcomes	There is strong support for SSC systems improving safety for both drivers and micromobility users. The "halo effect" remains a debated topic despite recognition in the literature. Agencies should educate the public about the halo effect and consider dynamic placement strategies for system-wide safety benefits.	Educate the public on the safety benefits of SSCs, addressing concerns about potential negative outcomes like increased speeding in non-monitored areas.
Location Guidance	There is public support for SSC deployment in high-risk areas such as school zones, work zones, police-limited areas, and community-nominated locations. While data-driven placement is essential, community involvement in site selection is also valued. Potential conflicts may arise between community preferences and efforts to mitigate the halo effect.	Use survey data to guide SSC placement, ensuring alignment with public expectations and safety needs.
Existing Conditions and Public Beliefs	SSC systems are perceived as effective for speed control on both high-speed and local roadways. There is general acceptance of mobile or hidden SSC systems as a solution to the halo effect. There is no strong consensus on whether facial verification for SSC enforcement constitutes a privacy invasion.	Address public concerns through transparent communication and education campaigns. Highlight the role of SSCs in improving road safety.
Transparency and Accountability	There is overwhelming public support for transparency regarding the operating agency, public reporting on system performance and citations, and clear revenue dissemination. Notification before systems become operational is also supported. Transparency must be a foundational principle for SSC system implementation.	Ensure transparency in SSC operations and data reporting. Provide regular updates to the public on program outcomes and revenue allocation.
Ticketing and Penalties	There is support for higher fines in certain areas, such as work zones, but opposition to lowering speed thresholds for ticket issuance. There is a strong preference for offering educational courses as an alternative to citations. There is support for issuing tickets to the driver rather than the vehicle owner, but resistance to facial recognition technology for enforcement. The most commonly supported speed threshold for ticketing is 10 mph over the posted speed limit.	Develop clear guidelines for ticket issuance, focusing on fairness and consistency. Use warnings and educational measures to complement enforcement.
Revenue Allocation	There is strong agreement that SSC-generated revenue should fund infrastructure repairs, safety improvements on high-speed and high-crash corridors, and law enforcement and micromobility enhancements. There are mixed opinions on whether SSC systems are primarily for safety improvements or revenue generation.	Allocate SSC revenue to safety projects and community improvements, particularly in underserved areas. Communicate revenue use to the public to build trust and support.

References

Abouchacra, Z., 2023. Automated Speed Enforcement as a Mechanism of Social Control? Ph.D. Thesis. Université d'Ottawa/University of Ottawa.

Abu-Rasheed, H., Weber, C., Fathi, M., 2024. Knowledge graphs as context sources for LLM-based explanations of learning recommendations. In: 2024 IEEE Global Engineering Education Conference (EDUCON). IEEE, pp. 1–5.

Afandizadeh, S., Sedighi, F., Kalantari, N., Mirzahossein, H., 2024. Modeling of the effect of transportation system accessibility on residential real estate prices: the case of washington metropolitan area, USA. Case Stud. Transp. Policy 18, 101277.

Afrose, S., 2024. Retrieval-augmented generation based doctor recommendation system using knowledge graph. Ph.D. Thesis. Brac University.

Ammar, A., Koubaa, A., Nacar, O., Boulila, W., 2025. Optimizing retrieval-augmented generation: analysis of hyperparameter impact on performance and efficiency. arXiv preprint arXiv:2505.08445.

Bao, K., Zhang, J., Zhang, Y., Wenjie, W., Feng, F., He, X., 2023. Large language models for recommendation: progresses and future directions. In: Proceedings of the Annual International ACM SIGIR Conference on Research and Development in Information Retrieval in the Asia Pacific Region, pp. 306–309.

Baur, D., Ansorg, J., Heyde, C.-E., Voelker, A., 2025. Development and evaluation of a retrieval-augmented generation chatbot for orthopedic and trauma surgery patient education: mixed-methods study. JMIR AI 4, e75262.

Beaton, M.D., Oakey, M., Newhouse, E., Copley, T.T., Fyfe, M., Karbakhsh, M., Turcotte, K., Zheng, A., Pike, I., 2022. Critical elements of public acceptance and support for automated speed enforcement in british columbia, canada. J. Transp. Health 26, 101461.

Chen, M., Tao, Z., Tang, W., Qin, T., Yang, R., Zhu, C., 2024. Enhancing emergency decision-making with knowledge graphs and large language models. Int. J. Disast. Risk Reduct. 113, 104804.

Chen, Y.-C., 2017. A tutorial on kernel density estimation and recent advances. Biostat. Epidemiol. 1 (1), 161–187.

Cui, Z., Weng, Y., Tang, X., Lyu, F., Liu, D., He, X., Ma, C., 2025. Comprehending knowledge graphs with large language models for recommender systems. In: Proceedings of the 48th International ACM SIGIR Conference on Research and Development in Information Retrieval, pp. 1229–1239.

Debinski, B., Smith, K.C., Gielen, A., 2014. Public opinion on motor vehicle–related injury prevention policies: a systematic review of a decade of research. Traffic Inj. Prev. 15 (3), 243–251.

Delbosc, A., Muir, C., Ralph, K., Barajas, J.M., Johnson-Rodriguez, A., 2025. Do perceptions of speeding act as a barrier to automated speed enforcement in the united states? Transp. Res. F: Traffic Psychol. Behav. 114, 1042–1052.

Deldjoo, Y., He, Z., McAuley, J., Korikov, A., Sanner, S., Ramisa, A., Vidal, R., Sathiamoorthy, M., Kasrizadeh, A., Milano, S., et al., 2024. Recommendation with generative models. arXiv preprint arXiv:2409.15173.

Ebrahim, Z., Nikraz, H., 2014. Camera system deployment for speeding control in australia. Int. J. Transp. Sci. Technol. 3 (4), 353–364.

Egbendewe-Mondzozo, A., Higgins, L.M., Shaw, W.D., 2010. Red-light cameras at intersections: estimating preferences using a stated choice model. Transp. Res. Part A: Policy Pract. 44 (5), 281–290.

Eriksson, L., Garvill, J., Nordlund, A.M., 2008. Acceptability of single and combined transport policy measures: the importance of environmental and policy specific beliefs. Transp. Res. A: Policy Pract. 42 (8), 1117–1128.

Es, S., James, J., Anke, L.E., Schockaert, S., 2024. Ragas: automated evaluation of retrieval augmented generation. In: Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics: System Demonstrations, pp. 150–158.

Federal Highway Administration, National Highway Traffic Safety Administration, 2023. Speed Safety Camera Program and Planning Operations Guide: An Overview. Technical Report FHWA-HOP-24-063. U.S. Department of Transportation. Overview document introducing the 2023 FHWA/NHTSA Speed Safety Camera Program Planning and Operations Guide. https://highways.dot.gov/sites/fhwa.dot.gov/files/Speed%20Safety%20Camera%20Program%20Planning%20and%20Operations%20Guide%202023.pdf.

Federal Highway Administration, U., 2008. Speed-Enforcement Camera Systems Operational Guidelines. Technical Report. Federal Highway Administration, Office of Safety. Washington, D.C. FHWA-SA-07-027. https://highways.dot.gov/safety/rwd/resources/speed-enforcement-camera-systems-operational-guidelines.

Figliozzi, M., Wang, H., Semensky, S., Sedighi, F., et al., 2025. Improving Guidance for Speed Safety Camera. Technical Report. Oregon Department of Transportation.

Fleiter, J.J., Watson, B., 2012. Automated speed enforcement in australia: recent examples of the influence of public opinion on program sustainability. J. Australas. Coll. Road Saf. 23 (3), 59–66.

Harrison, R.L., 2010. Introduction to Monte Carlo simulation. In: AIP Conference Proceedings. Vol. 1204, p. 17.

Hu, W., McCartt, A.T., 2016. Effects of automated speed enforcement in Montgomery County, Maryland, on vehicle speeds, public opinion, and crashes. Traffic Inj. Prev. 17 (sup1), 53–58.

Job, R.F.S., 2022. Evaluations of speed camera interventions can deliver a wide range of outcomes: causes and policy implications. Sustainability 14 (3), 1765.

Jovanović, M., Campbell, M., 2023. Connecting AI: merging large language models and knowledge graph. Computer 56 (11), 103–108.

King, J., Zineddin, A., Thomas, E., Williams, K., 2024. Speed safety cameras: steering clear of potholes during adoption planning, design, and operation. Public Road. 88 (1), 29–34.

Laban, P., Fabbri, A.R., Xiong, C., Wu, C.-S., 2024. Summary of a haystack: a challenge to long-context LLMs and RAG systems. In: Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing, pp. 9885–9903.

Laban, P., Hayashi, H., Zhou, Y., Neville, J., 2025. LLMs get lost in multi-turn conversation. arXiv preprint arXiv:2505.06120.

LangChain Developers, 2024. Llmgraphtransformer - LangChain experimental graph transformer module. Accessed: 2025-07-28. https://python.langchain.com/api_reference/experimental/graph_transformers/langchain_experimental.graph_transformers.llm.LLMGraphTransformer.html.

Lee, J., Ahn, S., Kim, D., Kim, D., 2024. Performance comparison of retrieval-augmented generation and fine-tuned large language models for construction safety management knowledge retrieval. Autom. Constr. 168, 105846.

Leonard, E., Sheehan, B., Mullins, M., Shannon, D., 2026. Generative AI for enhanced risk management in SMEs. J. Risk Res. , 1–34.

Li, L., Zhang, Y., Liu, D., Chen, L., 2023. Large language models for generative recommendation: a survey and visionary discussions. arXiv preprint arXiv:2309.01157.

Li, Y., Kim, A.M., El-Basyouny, K., Li, R., 2016. Using GIS to interpret automated speed enforcement guidelines and guide deployment decisions in mobile photo enforcement programs. Transp. Res. A: Policy Pract. 86, 141–158.

Lin, J., Dai, X., Xi, Y., Liu, W., Chen, B., Zhang, H., Liu, Y., Wu, C., Li, X., Zhu, C., et al., 2025. How can recommender systems benefit from large language models: a survey. ACM Trans. Inf. Syst. 43 (2), 1–47.

Lyon, C., Delavary, M., Cicchino, J., Hu, W., Wolfe, D., 2024. Examining Automated Speed Enforcement Programs in Canada.

McCarty, A., 2023. Use of Speed Camera Enforcement Systems in the United States.

Mehdizadeh, M., Solbu, G., Klöckner, C.A., Skjølsvold, T.M., 2024. Navigating acceptance and controversy of transport policies. Transp. Res. A: Policy Pract. 187, 104176.

Meng, Z., Yi, Z., Ounis, I., 2025. KERAG_R: knowledge-enhanced retrieval-augmented generation for recommendation. arXiv preprint arXiv:2507.05863.

Miller, J.J., 2013. Graph database applications and concepts with Neo4J. In: Proceedings of the Southern Association for Information Systems Conference, Atlanta, GA, USA. Vol. 2324, pp. 141–147.

Morain, S.R., Gielen, A.C., Bhalla, K., 2016. Automated speed enforcement systems to reduce traffic-related injuries: closing the policy implementation gap. Inj. Prev. 22 (1), 79–83.

Onuean, A., Lee, D., Jung, H., 2020. Traffic Safety Recommendation using Combined Accident and Speeding Data.

Oregon Legislative Assembly, J., 2024. House Bill 4109: Relating to Transportation; Prescribing an Effective Date. Introduced bill. 82nd Oregon Legislative Assembly, Oregon State Legislature. Introduced version - digest published in February 2024 session :contentReference[oaicite:1]index=1. https://olis.oregonlegislature.gov/liz/2024R1/Downloads/MeasureDocument/HB4109/Introduced.

Peterson, C., Douma, F., Morris, N., 2017. Addressing key concerns regarding automated speed enforcement via interactive survey. Transp. Res. Rec. 2660 (1), 66–73.

Pilkington, P., Kinra, S., 2005. Effectiveness of speed cameras in preventing road traffic collisions and related casualties: systematic review. Bmj 330 (7487), 331–334.

Pires, D., Perezhohin, Y., Castelli, M., 2025. Retrieval-augmented generation assistant for anatomical pathology laboratories. Emerg. Sci.

Potluri, D., Mathew, A.M., Rasgon, A.L., Dewitt, J.B., Hao, Y., McGrath, J.C., Hong, J., Nemeroff, C.B., Muller, G., Ding, Y., 2025. Position: CARE-RAG: clinical assessment and reasoning in RAG. In: The Second Workshop on GenAI for Health: Potential, Trust, and Policy Compliance.
Qiu, Z., Luo, L., Pan, S., Liew, A. W.-C., 2024. Unveiling user preferences: a knowledge graph and LLM-driven approach for conversational recommendation. arXiv preprint arXiv:2411.14459.
Reimers, N., Gurevych, I., 2019. Sentence-BERT: sentence embeddings using siamese BERT-networks. arXiv preprint arXiv:1908.10084.
Retting, R.A., Farmer, C.M., McCartt, A.T., 2008. Evaluation of automated speed enforcement in Montgomery County, Maryland. Traffic Inj. Prev. 9 (5), 440–445.
Rodier, C.J., Shaheen, S.A., Cavanagh, E., 2007. Automated speed enforcement in the US: a review of the literature on benefits and barriers to implementation. In: Transportation Research Board 87th Annual Meeting. CD-ROM. Washington, DC.
Rothman, D., 2024. RAG-Driven Generative AI: Build custom retrieval Augmented Generation Pipelines with LlamaIndex, Deep Lake, and Pinecone. Packt Publishing Ltd.
Roychowdhury, S., Soman, S., Ranjani, H.G., Gunda, N., Chhabra, V., Bala, S.K., 2024. Evaluation of RAG metrics for question answering in the telecom domain. arXiv preprint arXiv:2407.12873.
Saeidian, P., Singleton, P., 2025. Trip production models comparing persons with and without disabilities. In: International Conference on Transportation and Development 2025, pp. 945–960.
Sar, A., Choudhury, T., Sati, S., Joshi, P., Aich, S., Pant, B., Dewangan, B.K., 2024. Based chatbot development using LLAMA2 and LangChain: training and deployment for document interaction. In: 2024 OPJU International Technology Conference (OTCON) on Smart Computing for Innovation and Advancement in Industry 4.0. IEEE, pp. 1–6.
Schechtman, E., Bar-Gera, H., Musicant, O., 2016. Driver views on speed and enforcement. Accid. Anal. Prev. 89, 9–21.
Sharifi, D., Mirzahossein, H., Kalantari, N., Sedighi, F., 2026. Integrating hybrid recurrent neural networks and large language models for daily travel behavior prediction. Transp. Res. Interdiscip. Perspect. 35, 101793.
da Silva, T.C., 2025. Extracting knowledge graphs from user stories using LangChain. arXiv preprint arXiv:2506.11020.
Sivakumar, A., Sugumaran, V., Qiang, Y., 2026. Rag-x: systematic diagnosis of retrieval-augmented generation for medical question answering. arXiv preprint arXiv:2603.03541.
Soole, D., Lennon, A., Watson, B., 2008. Driver perceptions of police speed enforcement: differences between camera-based and non-camera based methods: results from a qualitative study. In: Road Safety 2008: Safer Roads, Safer Speeds, Safer People, Safer Vehicles-proceedings of the 2008 Australasian Road Safety Research, Policing and Education Conference. Causal Productions, pp. 221–231.
Tapp, A., Nancarrow, C., Davis, A., Jones, S., 2016. Vicious or virtuous circles? Exploring the vulnerability of drivers to break low urban speed limits. Transp. Res. A: Policy Pract. 91, 195–212.
Thakur, N., Reimers, N., Rücklé, A., Srivastava, A., Gurevych, I., 2021. Beir: a heterogenous benchmark for zero-shot evaluation of information retrieval models. arXiv preprint arXiv:2104.08663.
Thomas, L.J., Srinivasan, R., Decina, L.E., Staplin, L., 2008. Safety effects of automated speed enforcement programs: critical review of international literature. Transp. Res. Rec. 2078 (1), 117–126.
Tian, J., Wang, Z., Zhao, J., Ding, Z., 2024. Mmrec: LLM based multi-modal recommender system. In: 2024 19th International Workshop on Semantic and Social Media Adaptation & Personalization (SMAP). IEEE, pp. 105–110.
Tilahun, N., 2023. Safety impact of automated speed camera enforcement: empirical findings based on Chicago's speed cameras. Transp. Res. Rec. 2677 (1), 1490–1498.
Topsakal, O., Akinci, T.C., 2023. Creating large language model applications utilizing LangChain: a primer on developing LLM apps fast. In: International Conference on Applied Engineering and Natural Sciences. Vol. 1, pp. 1050–1056.
Vats, A., Jain, V., Raja, R., Chadha, A., 2024. Exploring the impact of large language models on recommender systems: an extensive review. arXiv preprint arXiv:2402.18590.
Wang, M., Guo, Y., Zhang, D., Jin, J., Li, M., Schonfeld, D., Zhou, S., 2024a. Enabling explainable recommendation in e-commerce with LLM-powered product knowledge graph. arXiv preprint arXiv:2412.01837.
Wang, Q., Li, J., Wang, S., Xing, Q., Niu, R., Kong, H., Li, R., Long, G., Chang, Y., Zhang, C., 2024b. Towards next-generation llm-based recommender systems: a survey and beyond. arXiv preprint arXiv:2410.19744.
Xu, K., Zheng, H., Zhan, X., Zhou, S., Niu, K., 2024a. Evaluation and optimization of intelligent recommendation system performance with cloud resource automation compatibility. Appl. Comput. Eng. 87, 228–233.
Xu, S., Hua, W., Zhang, Y., 2024b. Openp5: an open-source platform for developing, training, and evaluating llm-based recommender systems. In: Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval, pp. 386–394.
Yamamoto, V.E., Kabal, O., Karunathilake, L., Nishigori, K., Lermanda, V., Zhao, S., Uematsu, H., He, Y., Takeda, H., 2025. Exploring LLM to Extract Knowledge Graph from Academic Abstracts .
Yu, P., Xu, Z., Wang, J., Xu, X., 2025. The application of large language models in recommendation systems. In: International Conference on Artificial Intelligence and Machine Learning Research (CAIMLR 2024). Vol. 13635. SPIE, pp. 103–112.
Zhang, Q., Chen, S., Bei, Y., Yuan, Z., Zhou, H., Hong, Z., Dong, J., Chen, H., Chang, Y., Huang, X., 2025. A survey of graph retrieval-augmented generation for customized large language models. arXiv preprint arXiv:2501.13958.
Zhao, Z., Fan, W., Li, J., Liu, Y., Mei, X., Wang, Y., Wen, Z., Wang, F., Zhao, X., Tang, J., et al., 2024. Recommender systems in the era of large language models (LLMs). IEEE Trans. Knowl. Data Eng. 36 (11), 6889–6907.
Zheng, Q., Sharmeen, F., Xu, C., Liu, P., 2024. Assessing regional road traffic safety in sweden through dynamic panel data analysis: influence of the planned innovative policies and the unplanned COVID-19 pandemic. Transp. Res. A: Policy Pract. 179, 103918.